



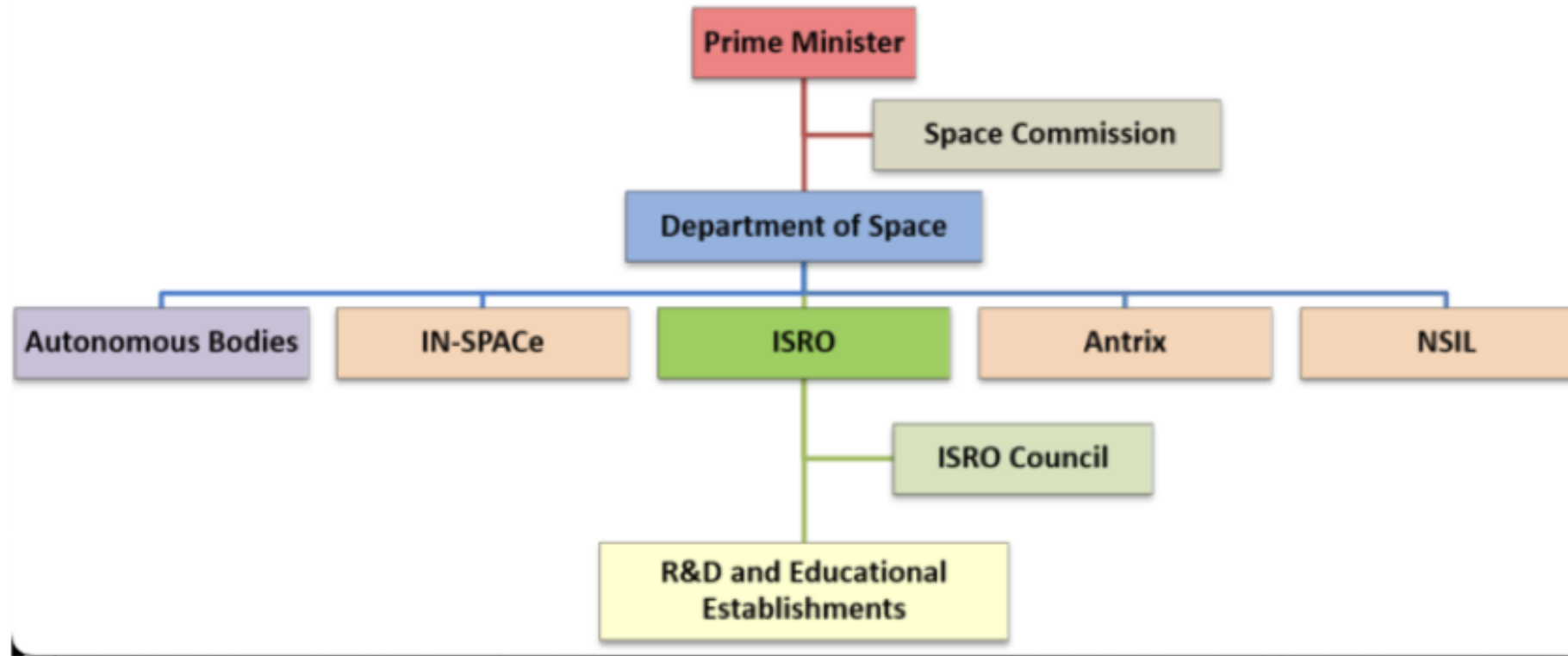
UPSC Prelims

Science & Tech

Class No. 10

Space Technology continued

Indian Space Architecture



ANTRIX

- Antrix—>Antrix Corporation Limited (ACL) is a PSU
- It is a wholly owned Government of India Company under the administrative control of the Department of Space.
- It was incorporated as a private limited company owned by Government of India in September 1992.
- It's mandate was to act as Marketing arm of ISRO for promotion and commercial exploitation of space products, technical consultancy services and transfer of technologies developed by ISRO.
- Another major objective is to facilitate development of space related industrial capabilities in India.
- As the commercial and marketing arm of ISRO, Antrix is engaged in providing Space products and services to international customers worldwide.

NSIL

- It was founded in 2019, by Department of Space (DOS) to bridge gap between ISRO and private sector and facilitates transfer of ISRO technologies to industry.
- It is a wholly-owned Government of India company, under the administrative control of DOS.
- It will not replace ANTRIX, a private limited company owned by Government of India in 1992 as a Marketing arm of ISRO

NSIL

- Central PSU under DoS
- Antrix will handle ISRO's commercial deals for satellites and launch vehicles with foreign customers.
- NSIL will deal with capacity building of local industry for space manufacturing.
- Vision : build launch vehicles, providing launch services, build satellites, providing space-based services, technology transfers,
- space-related products and services to global customers
- spur the growth of Indian Industry in undertaking technologically challenging space-related activities

NSIL

- NSIL's main purpose is to market the technologies developed by ISRO and bring it more clients that need space-based services.
- That role, incidentally, was already being performed by Antrix Corporation, another PSU working under the Department of Space, and which still exists.

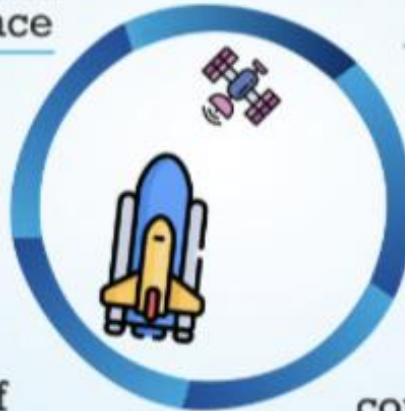


NEW SPACE INDIA LIMITED



NSIL - a Public Sector Enterprise - incorporated as new commercial arm of Department of Space

To harness India's space power commercially



To tap the benefits of R & D carried out by ISRO

To spearhead commercialization of space products including production of launch vehicles, transfer to technologies and marketing of space products

Private Sector in Space

- Government has created Indian National Space Promotion and Authorization Centre (INSPACe) to boost private sector participation in entire range of space activities.
- It is a single-window, independent, nodal agency which functions as an autonomous agency in the Department of Space (DOS).
- It is a single window agency for all space sector activities of private entities, IN-SPACe plays an important role in boosting the private space sector economy in India.



i.


Promote
and develop Indian space ecosystem accelerating space economy.


Enable
and nurture NGEs to accomplish their ventures in space sector.


Authorize
space operations and services in the country through well defined framework.


Supervise
the space activities of NGEs in the country



INDIAN NATIONAL SPACE PROMOTION AND AUTHORISATION CENTRE (IN-SPACE)


Ahmedabad, Gujarat

About: Autonomous nodal agency under **Department of Space** to promote, authorize and supervise **space activities of Non-Governmental Private Entities in India**.



Composition: A chairman, technical experts for space activities, safety experts, experts from Academia and Industries, members from **Prime Minister's Office** and **Ministry of External Affairs**.



Mandate:

- ⇨ **Sharing of space infrastructure** and premises under the control of ISRO.
- ⇨ **Establishment of temporary facilities** within premises under ISRO.
- ⇨ **Establishment of new space infrastructure and facilities.**



Other Key Information: Decision of IN-SPACE in context of this mandate, shall be final and binding on all stakeholders including ISRO.

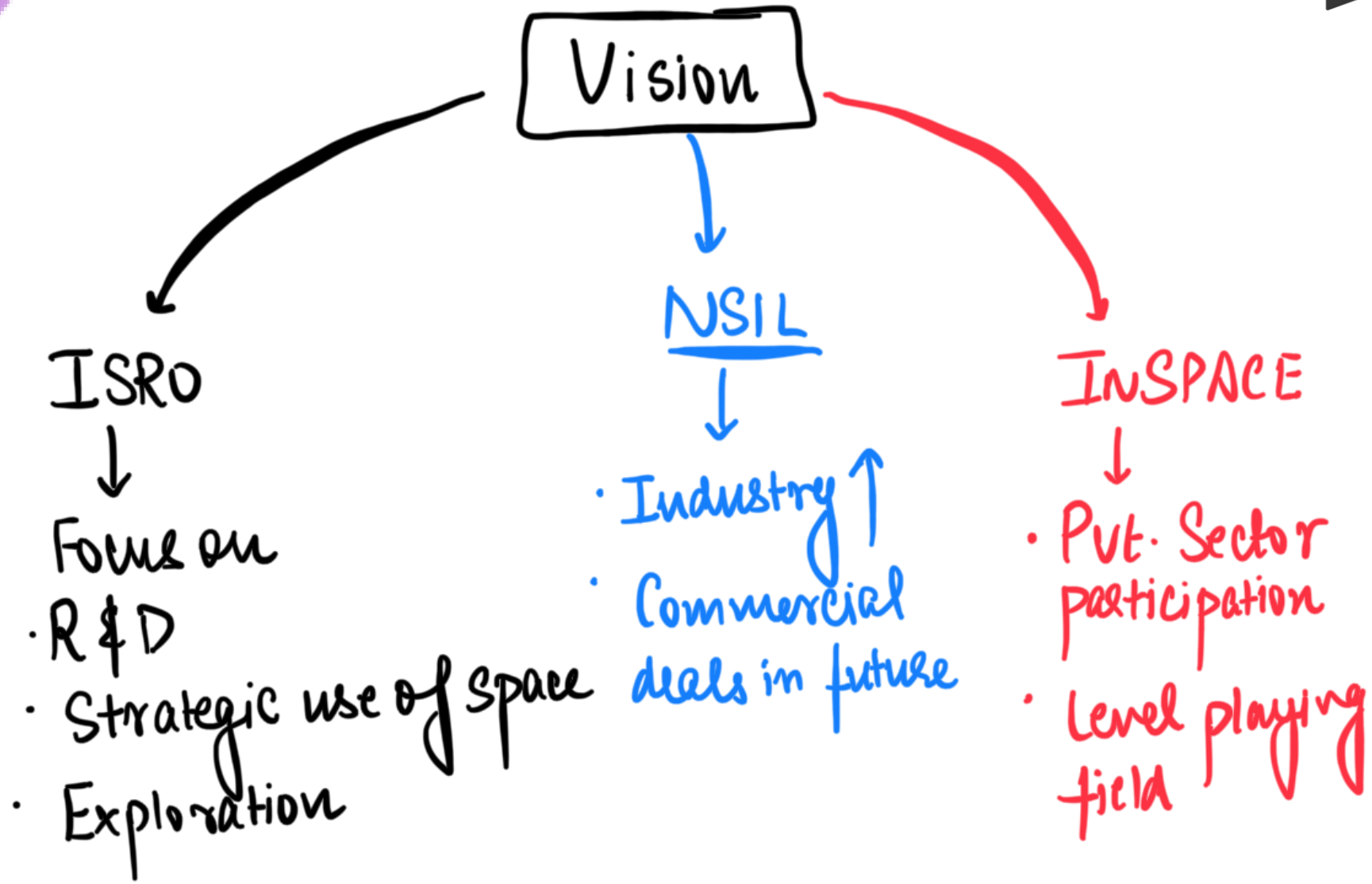


Private Sector in Space

- Interface—>agency acts as an interface between ISRO and Non-Governmental Entities (NGEs) and assesses how to utilize India's space resources better and increase space-based activities.
- Assess—>It also assesses the needs and demands of private players, including educational and research institutions, and explore ways to accommodate these requirements in consultation with ISRO.
- Accessibility—>Existing ISRO infrastructure, both ground- and space-based, scientific, and technical resources, and even data can be made accessible to interested parties to enable them to carry out their space-related activities through IN-SPACe.

Key Differences

- Antrix will handle ISRO's commercial deals for satellites and launch vehicles with foreign customers.
- NSIL will deal with capacity building of local industry for space manufacturing.
- The main objective of NSIL is to scale up industry participation in Indian space programmes in comparison to IN-SPACe which gives emphasis on the participation of the private sector.
- **However, NSIL is ALSO doing commercial deals.**

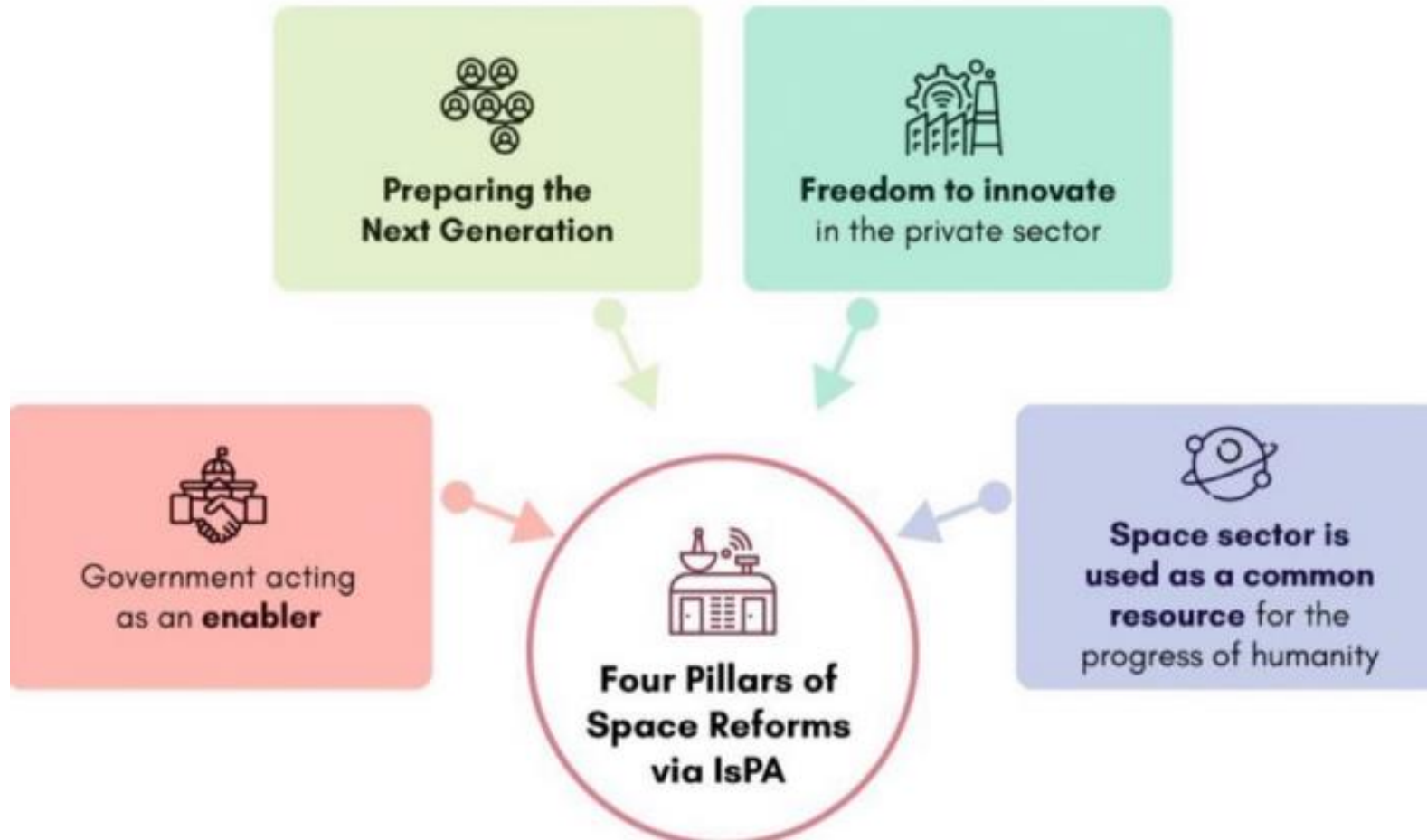


Indian Space Association

- It is an industry body consisting of various stakeholders of the Indian space domain.
- The industry association will act as an independent and a “single- window” agency for enabling the opening up of the space sector to start-ups and the private sector.
- ISpA will also work towards building global linkages for the Indian space industry to bring in critical technology and investments into the country to create more high-skill jobs

Indian Space Association

- The current global space economy is estimated at US\$ 360 billion, with India accounting for ~2% (US\$ 7 billion).
- In the global space market, rocket and satellite launch services, an area in which ISRO specialises, amount to only 5% share. Satellite-based services and ground-based systems account for the remaining 95%.
- Several private companies are developing plans to take paying customers to space on a regular basis leading to a rapid growth in space tourism industry.



Recent initiatives by Private Sector

- Hyderabad's Dhruva Space and Bengaluru's Digantara were authorized by IN-SPACe for launch of payloads onboard PSLV Orbital Experimental Module (POEM) of PSLV-C53.
- Mission Prarambh: Involved launch of Vikram-S (VKS), India's first privately built rocket by Hyderabadbased Skyroot Aerospace.
- VKS is a single stage spin stabilized solid propellant rocket where 3D printing has been used.
- Payload Capacity: 290 kg- 560 kg support from ISRO and IN-SPACe

Recent initiatives by Private Sector

- India's first private launch pad unveiled at Sriharikota
- Designed by Agnikul (a startup) and executed in support of ISRO and IN-SPACe.
- It is specifically built to support liquid-stage controlled launches.

Space Tourism

- Space tourism is space travel for recreational, leisure or business purposes. There are several different types of space tourism, including orbital, suborbital and lunar space tourism.
- To date, **orbital space tourism has been performed only by the Russian Aviation and Space Agency- Roscosmos.**
- Several private spaceflight companies are now working towards developing suborbital space tourism vehicles to take paying customers to space.

Space Tourism

- **No international space law has defined space tourists.** Existing space treaties such as Outer Space Treaty, Rescue Agreement etc. are only applicable to astronauts, envoys of mankind, or personnel of a spacecraft.
- **To send a vehicle to space, it is compulsory for the national and international space law to authorize it according to Article VI of the Outer Space Treaty.** However, there is lack of clarity on authorization of space travel with tourists aboard.

Space Travel

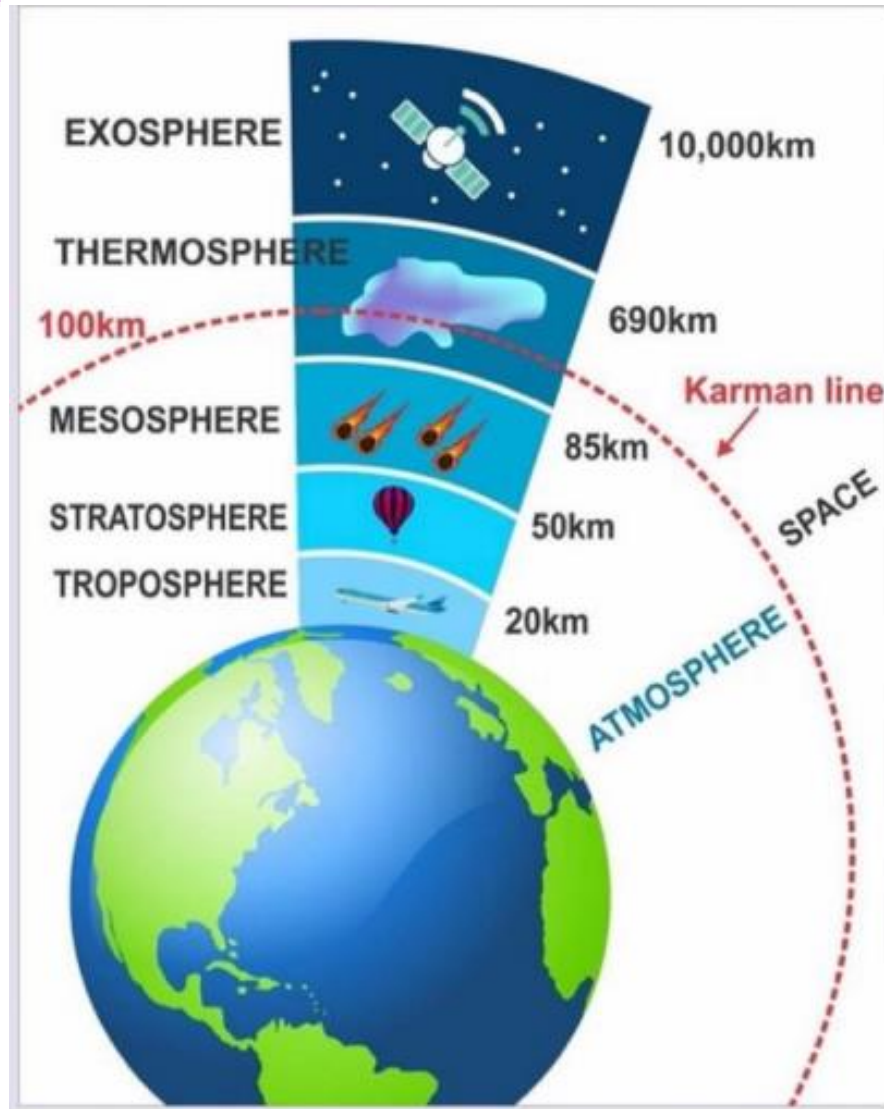
- According to the Fédération Aéronautique Internationale (world governing body for aeronautic and astronautic records), space starts at an altitude of 100 km (62 miles) above the surface of the earth.
- This is the Karman line where atmospheric lift no longer supports a flying object and the object would need to reach orbital velocity or risk falling back to Earth.

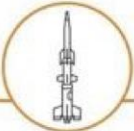
Space Travel

- Space travel is referred to as any flight operation that takes one or more passengers beyond the altitude of 100 km and thus into space.
- The main difference between orbital and suborbital flight is the speed at which a vehicle is traveling.
- An orbital spacecraft must achieve orbital velocity i.e. the speed that an object must maintain to remain in orbit around a planet. To orbit 125 miles (200 kilometers) above Earth for instance, a spacecraft must travel at a screaming 17,400 mph (28,000 km/h).

Space Travel

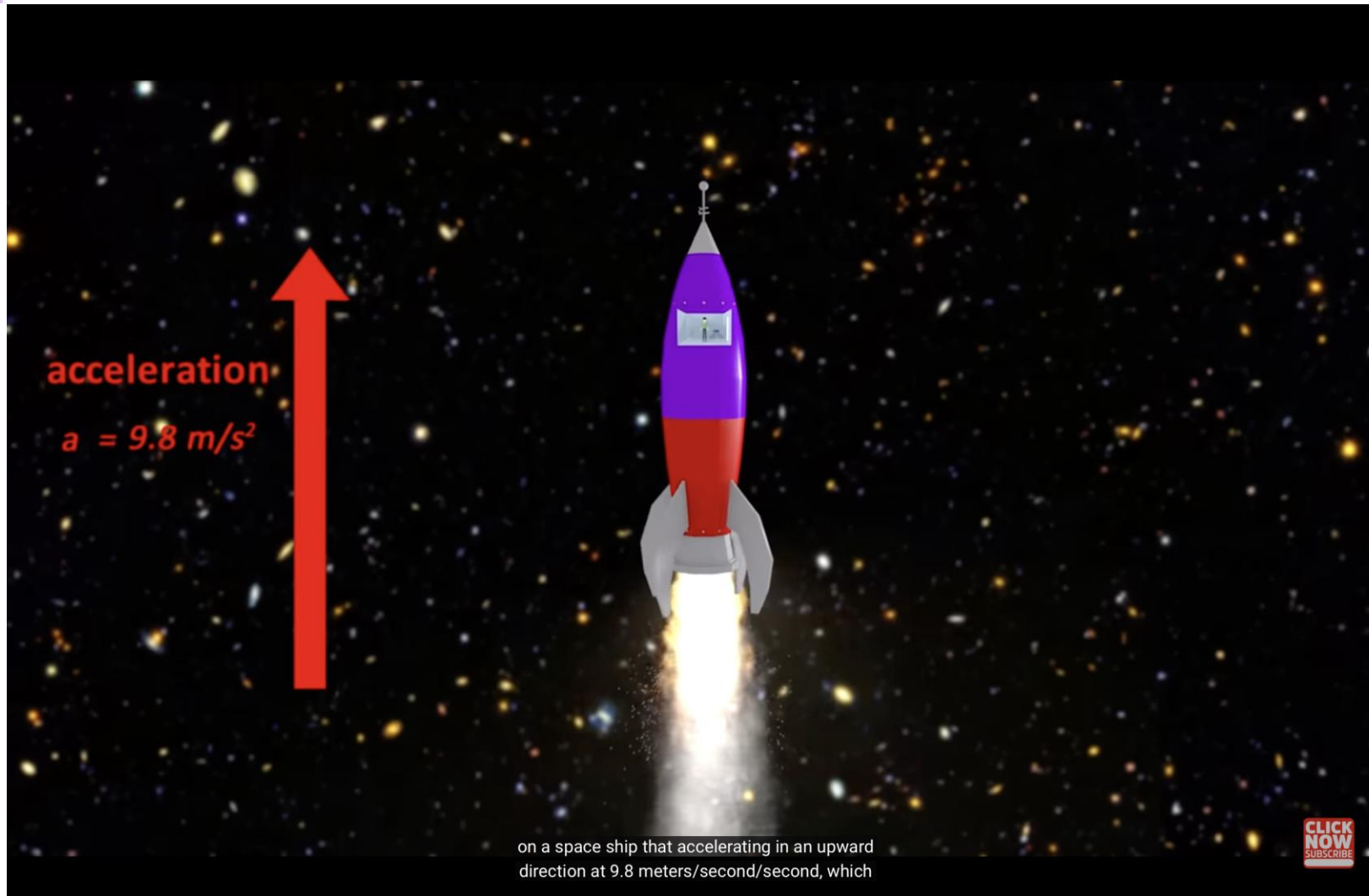
- Suborbital flight, in contrast, requires much lower speeds and doesn't have the power to achieve orbit. Instead, it will fly up to a certain height that depends on its speed, and then come back down once its engines are shut off.
- At the top of their flight arc in a suborbital flight, when the object is falling back toward Earth, passengers achieve a few minutes of weightlessness under zero gravity.
- Inspiration 4 : first orbital trip where none of the people aboard is a professional astronaut. (by space X)



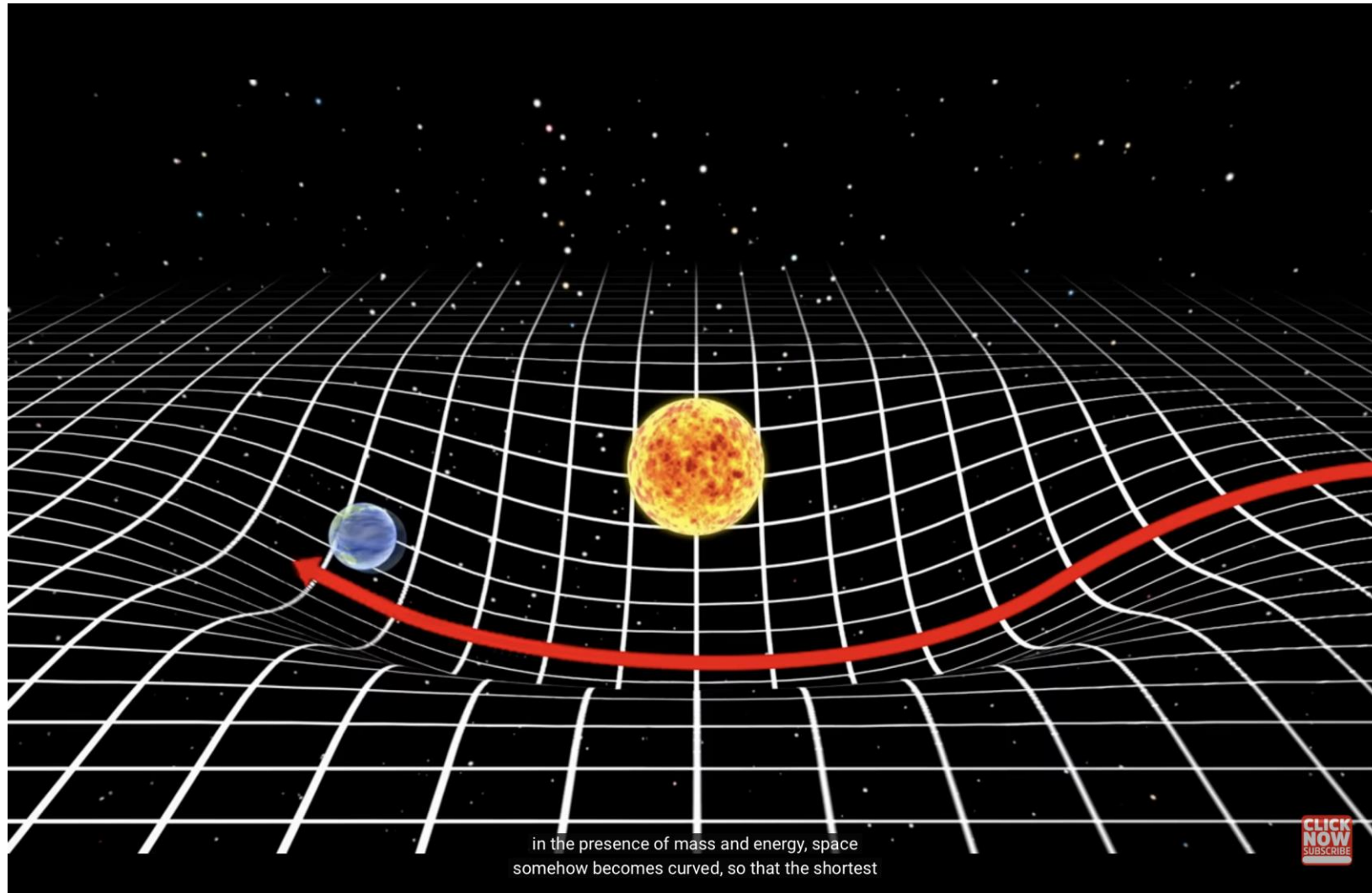
	 Suborbital tourism	 Orbital tourism
Altitude	About 100 kilometres	Over 400 kilometres
Duration in Space	Gives passengers a few minutes in space	Spend days or even more than a week in space.
Velocity	Sub-Orbital  • Requires much lower speeds (than orbital) and doesn't have the power to achieve orbit. • Instead, it will fly up to a certain height and then come back down once its engines are shut off.	Orbital  • Spacecraft must achieve orbital velocity i.e. the speed that an object must maintain to remain in orbit around a planet.

Black Hole

- Star Basics : Balance between radiation and gravity
- Radiation comes from nuclear fusion in the core
- Gravity we have already understood as a force : but now we will see it as bending of space time
- Let's understand these terms :







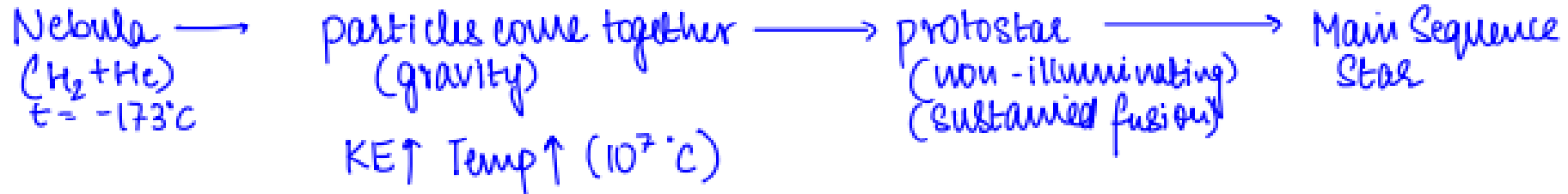
Let us see a video : how bending of space time explains gravitational motion

Links :

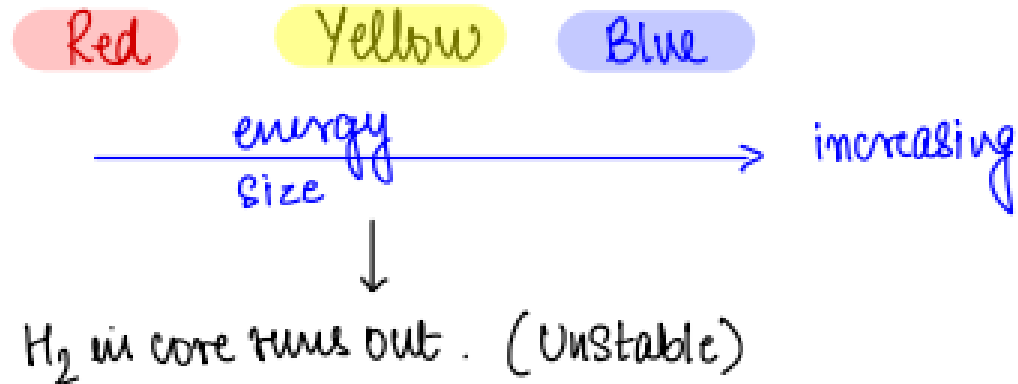
<https://www.youtube.com/watch?v=tzQC3uYL67U>

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What Happens to a Star?



Main Sequence Star: Core has fusion Rx^n



Black Holes

- Black hole is a place in space where gravity pulls so much that even light cannot get out. The gravity is so strong because matter has been squeezed into a tiny space.
- o This can happen when a big star is dying (our sun will never turn into a black hole as it is not big enough to make a black hole).
- o Because no light can get out, so they are invisible

Black Holes

- In the center of a black hole is a gravitational singularity, a one-dimensional point which contains a huge mass in an infinitely small space, where density and gravity become infinite and space-time curves infinitely, and the laws of physics as we know them cease to operate.

Black Holes

- Parts of Black hole :
- Accretion Disc
- Singularity
- Event Horizon
- Schwarzschild Radius : radius of event horizon
- ErgoSphere : Black hole is rotating, its mass causes the space time around black hole to rotate as well.

Black Holes

Singularity: the very centre of a black hole where matter has collapsed in a region of infinite density

Relativistic jet: when stars are absorbed by black holes, jets of particles and radiation are blasted out at near light speed

Photon sphere: photons emitted from hot plasma near the black hole which bends their trajectory producing a bright ring

Accretion disc of superheated gas and dust whirls around black hole at immense speeds, producing electromagnetic radiation (x-rays)

Event horizon: the radius around a singularity where matter and energy cannot escape the black hole's gravity. The point of no return.

Categories of Black Hole



Stellar

Formed by the gravitational collapse of a star. Masses ranging from about 5 to several tens of solar masses.



Supermassive

The largest type of black hole, containing a mass of the order of hundreds of thousands, to billions of times, the mass of the sun.



Intermediate-mass

Mass in range $10^2 - 10^5$ solar masses; significantly more than stellar black holes but less than $10^5 - 10^9$ solar mass supermassive black holes.



Miniture

Approximately 148 fm, Hypothetical tiny black holes, for which quantum mechanical efforts play an important role. The concept that black holes may exist that are smaller than stellar mass was introduced in 1971 by famous astrophysicist Stephen Hawking.

Black Holes

- Parts of Black hole :
- Accretion Disc
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- Schwarzschild Radius : radius of event horizon
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Taking Pictures of a black hole

- In 2019 scientists got the first optical image of a black hole through Event Horizon Telescope.
- It has captured the just outside region of a black hole, located 55 million light-years from Earth, at the centre of a galaxy named Messier 87. The image shows a photon (light quantum) can orbit the black hole without falling in. This is called the 'last photon ring'
- Sagittarius A* is the second black hole whose photographs have been captured by the Event Horizon Telescope project : at the center of milky way galaxy.

Taking Pictures of a black hole

- 2020 nobel in physics : Roger Penrose : black hole formation is based on general theory of relativity
- Reinhard Genzel and Andrea Ghez : discovery of supermassive compact object at the center of our galaxy : most likely a supermassive black hole : Sagittarius A*

Black holes and research

- Significance of black hole in space research
- Evolution of Galaxies
- Distribution of Dark Matter
- Star Formation
- Graviational Waves
- Testing general theory of relativity
- Light from behind the black hole : bending

Black holes and research

- Star formation: In particular, Supermassive black holes play an important role in star formation within galaxies.
- Gravitational waves: Scientists have detected that gravitational waves are generated when two black holes collide, and found that ringing pattern of the wave predicts the cosmic body's mass and spin.

Black holes and research

- General theory of relativity: The discovery of black hole has opened the door to several new lines of scientific investigation, enabling quantitative estimates of black hole related parameters. It has provided another laboratory to test the predictions of the Einstein's general theory of relativity.

Black holes and research

- Bending of light: The bending of light around a black hole is very significant because of its extremely large mass. Because of this bending, one can see light from material that is behind the black hole, which would not be seen if light were traveling in straight lines.

General theory of Relativity

- This theory was proposed by Albert Einstein in 1915.
- Essentially, it's a theory of gravity whose basic idea is that instead of being an invisible force that attracts objects to one another, gravity is a curving or warping of space. The more massive an object, the more it warps the space around it.
- The warping also affects measurement of time. Gravity can stretch space, it can also dilate time.

General theory of Relativity

- For example, the sun is massive enough to warp space across our solar system (a bit like the way a heavy ball resting on a rubber sheet warps the sheet). As a result, Earth and the other planets move in curved paths (orbits) around it.
- General theory of relativity explained the precession of Mercury

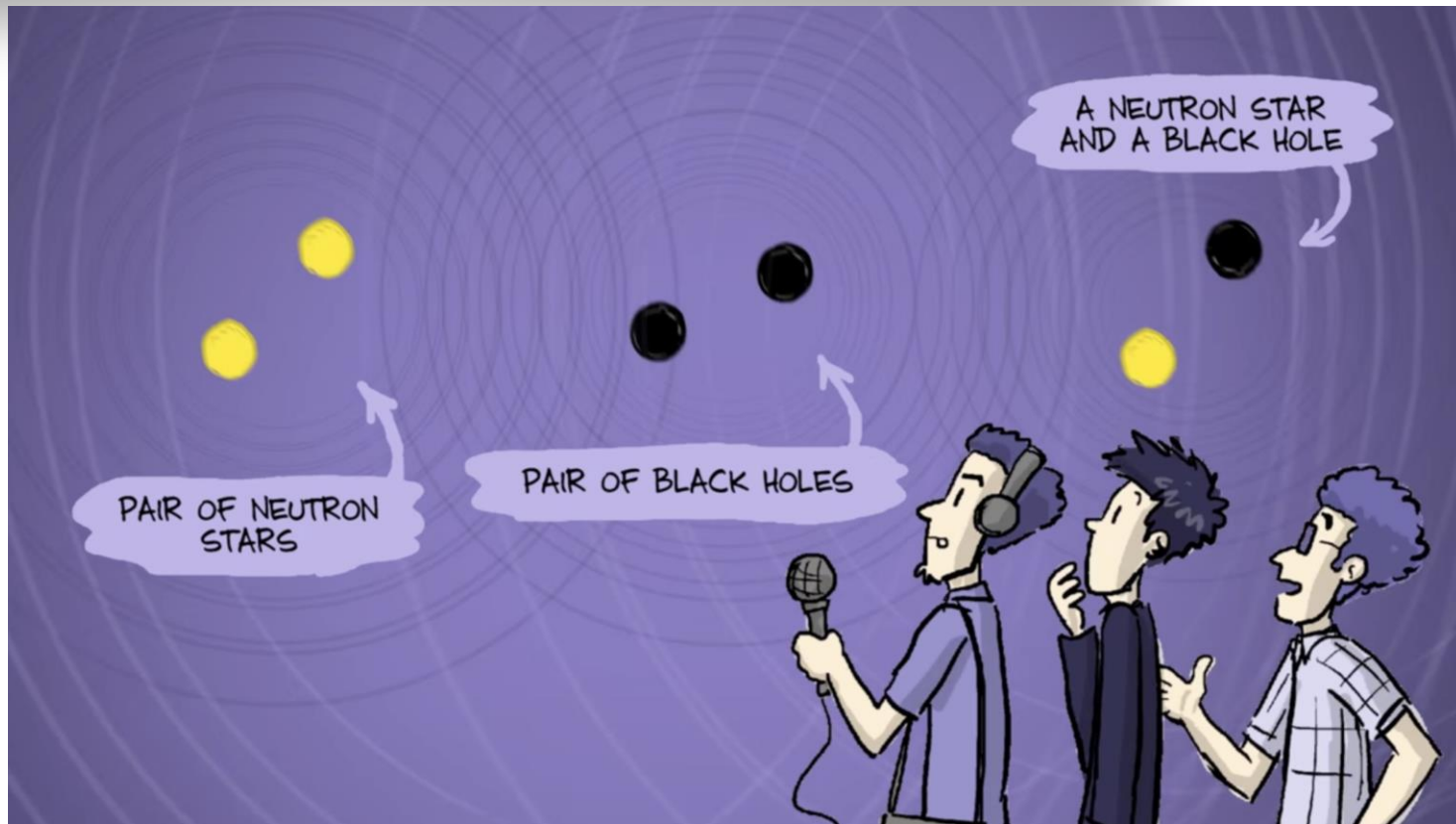
General theory of Relativity

- Confirmation : In the first major test of general relativity, astronomers in 1919 measured the deflection of light from distant stars as the starlight passed by our sun, proving that gravity does, in fact, distort or curve space.
- In 2016, the discovery of gravitational waves (subtle ripples in the fabric of spacetime) was another confirmation of general relativity.

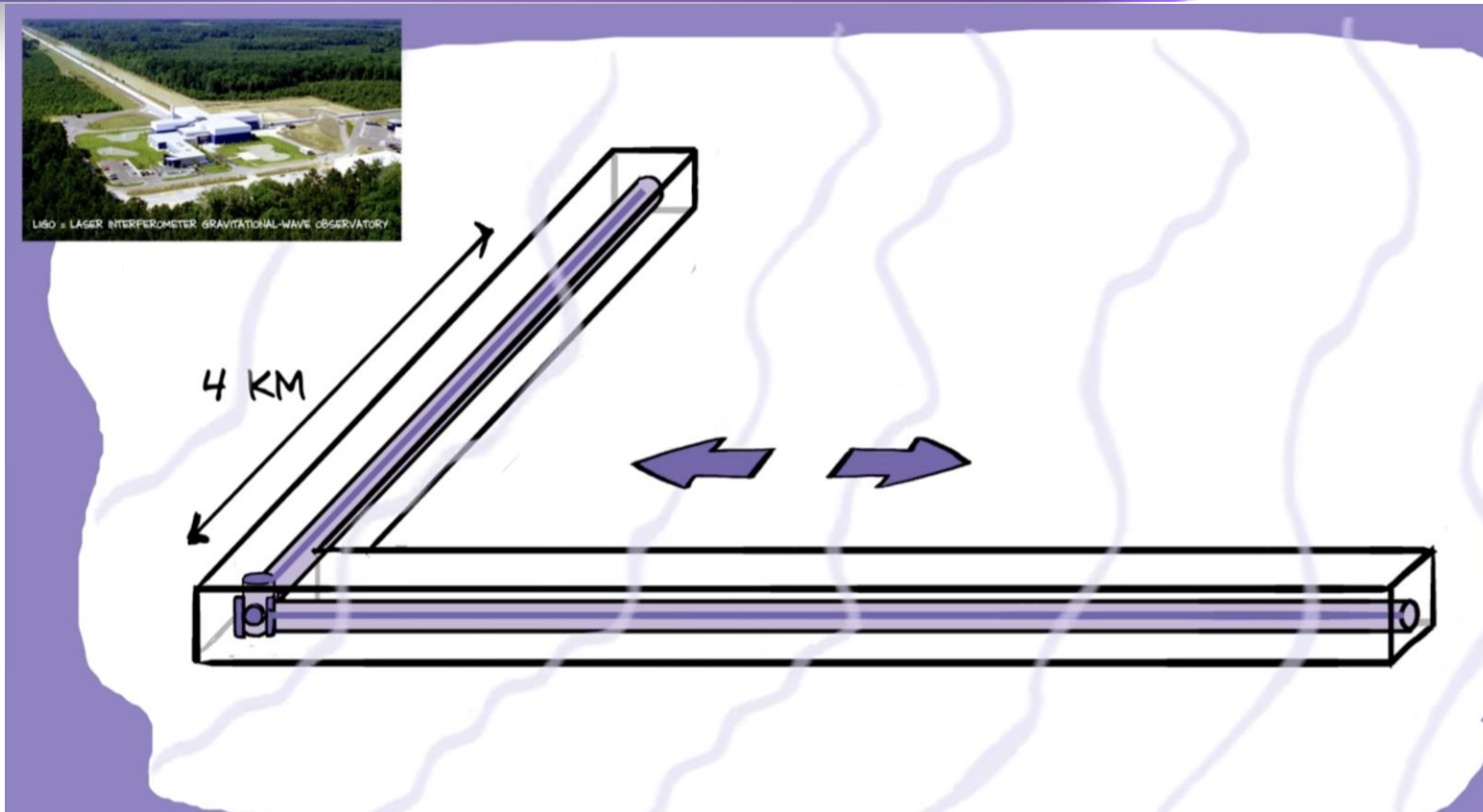
General theory of Relativity

- Gravitational waves are produced by cataclysmic events such as colliding black holes, supernovae (massive stars exploding at the end of their lifetimes), and colliding neutron stars. They travel at speed of light, squeezing and stretching anything in their path.
- Gravitational Waves were detected for the first time in the LIGO : Laser Interferometer Gravitational Wave Observatory

Gravitational Waves

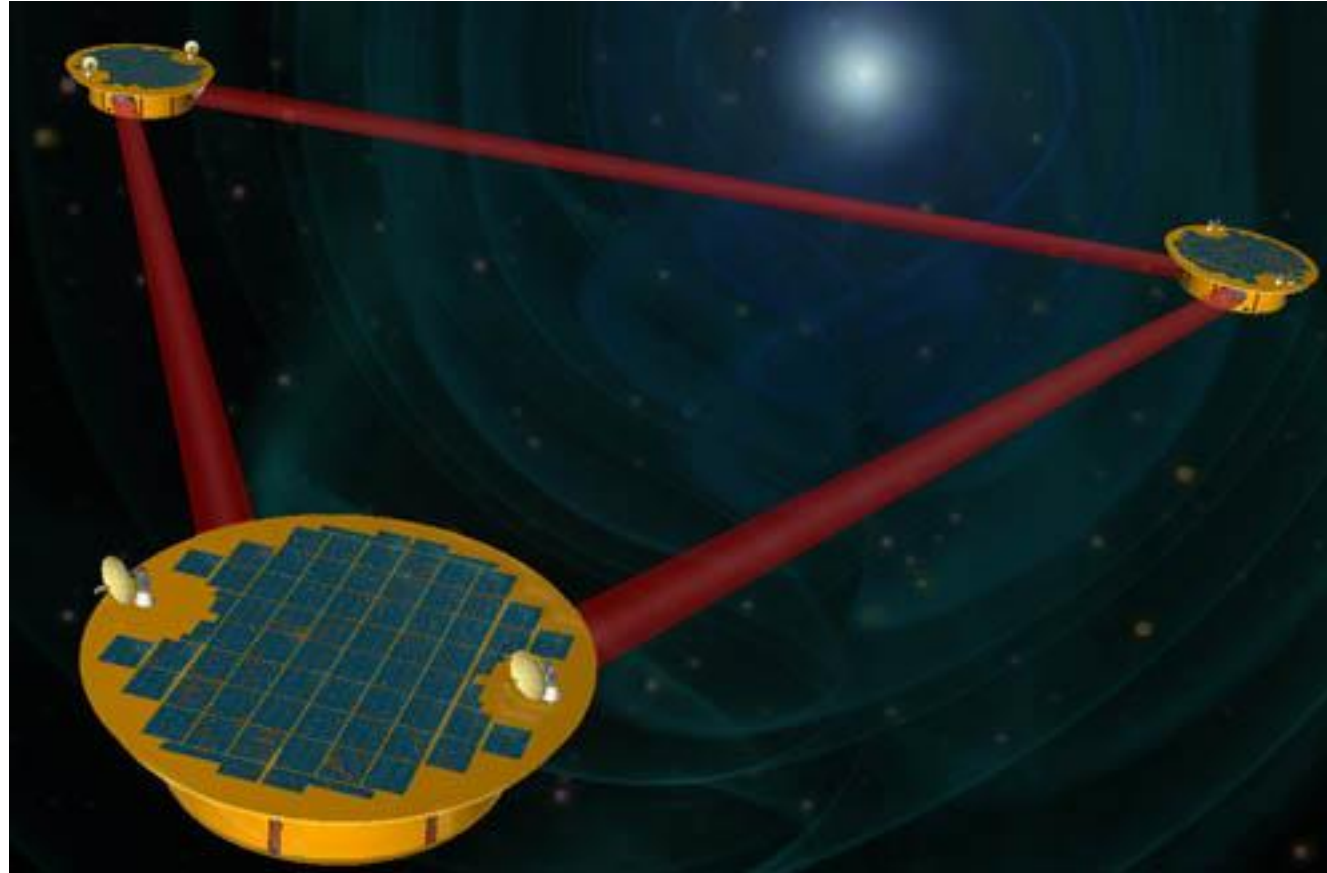


Gravitational Waves : LIGO



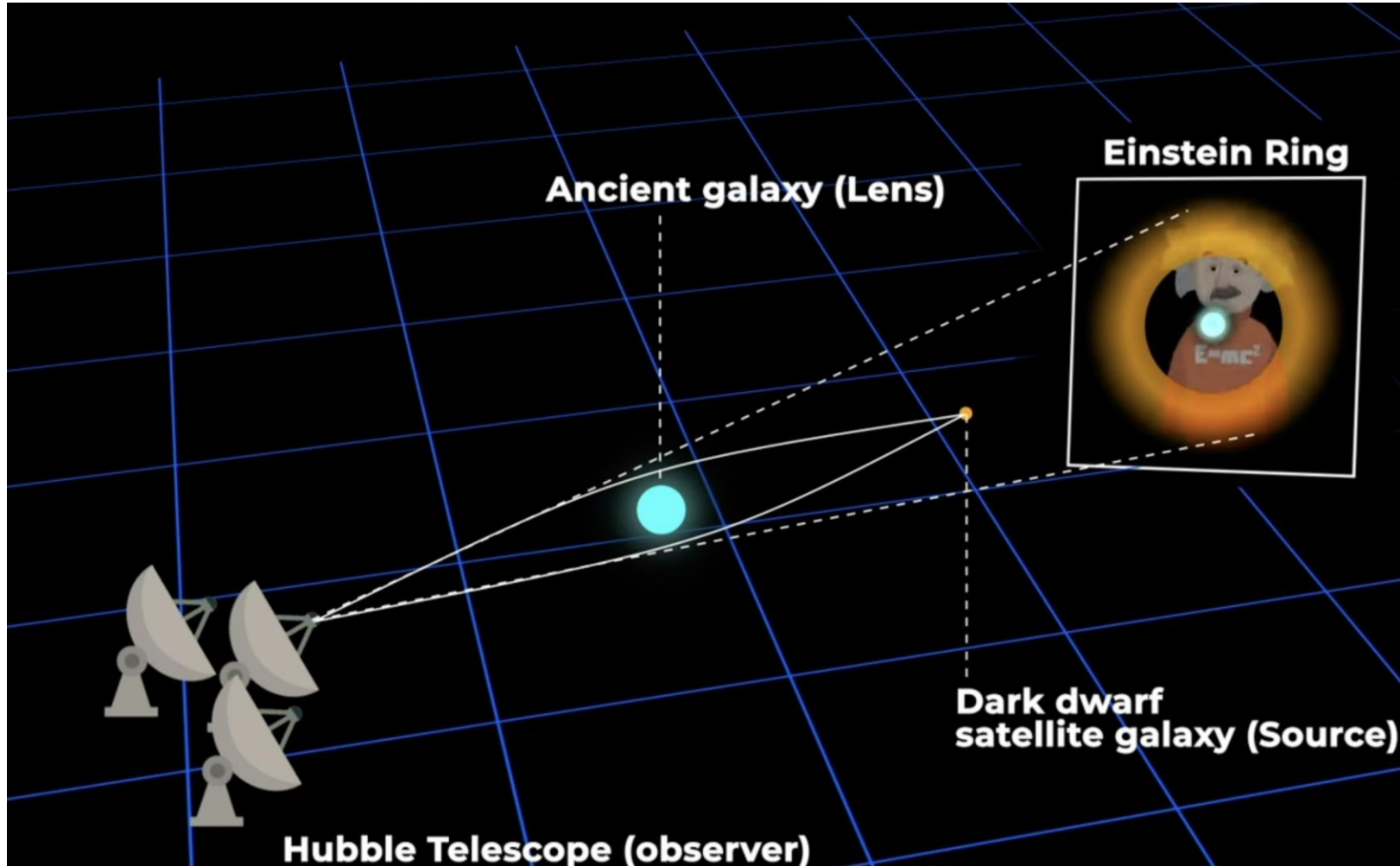
E-Lisa

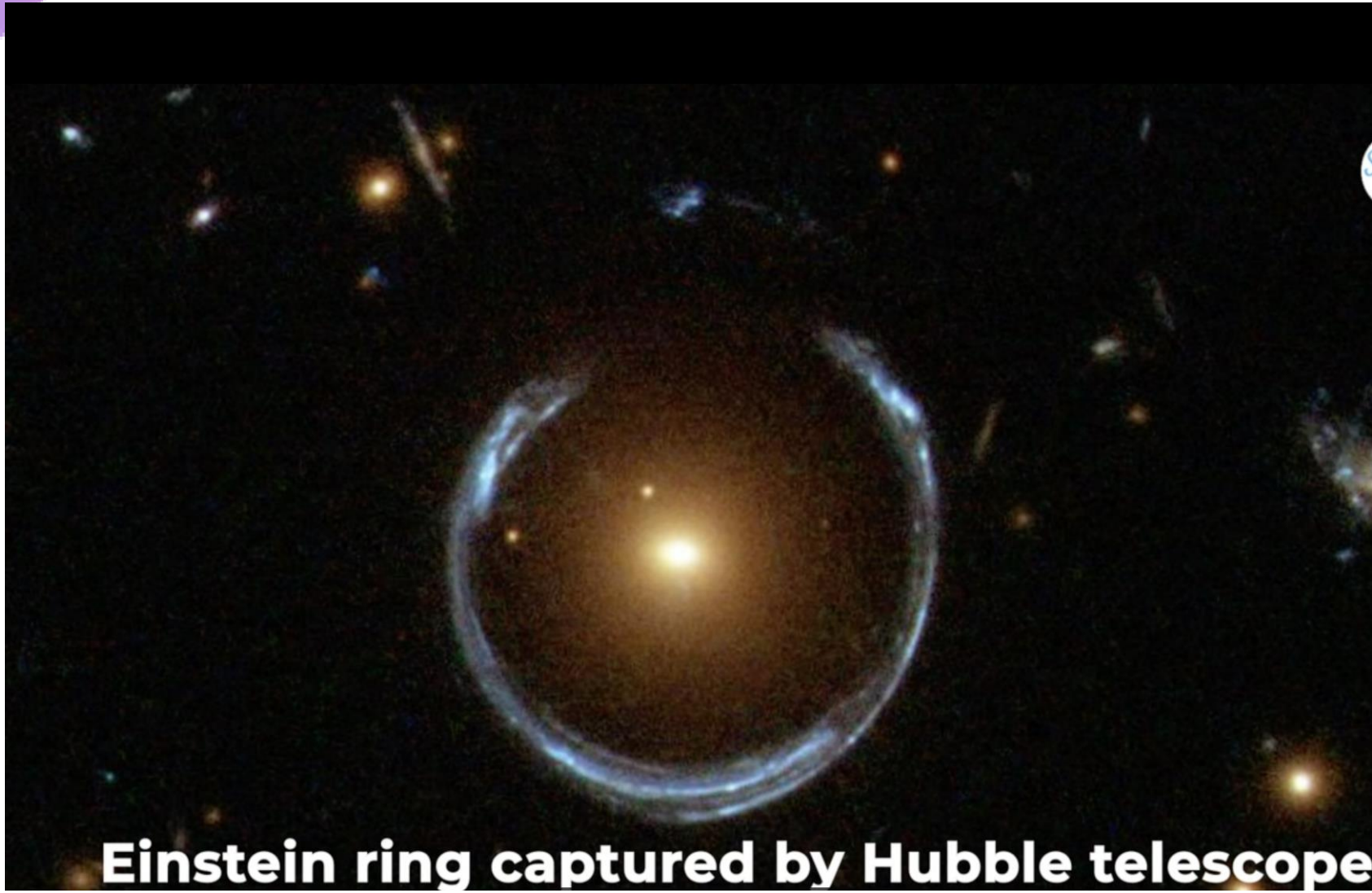
- eLISA's mission is to explore the Gravitational waves from space for the first time.
- The eLISA mission consists of a “Mother” and two “Daughter” spacecraft. These will orbit the Sun in a triangular configuration, as shown in the figure.
- The three satellites will form a precision interferometer, with the two Daughter spacecraft connected to the Mother one by 1 million km long laser beams.
- This interferometer will be capable of detecting gravitational waves at frequencies in the range of 0.1 mHz to 1 Hz.
- Such a frequency interval is not accessible on Earth due to arm length limitations and to noise caused by the terrestrial gravity gradient noise: in this sense, eLISA will complement the efforts of ground-based gravitational-wave detectors. Ex LIGO



Gravitational Lensing

- Lens : anything that distorts light rays to make it easier for us to see.
- When gravity distorts light rays it is called gravitational lensing : gravity leads to curving of light allowing us to see things that we wouldn't see normally.

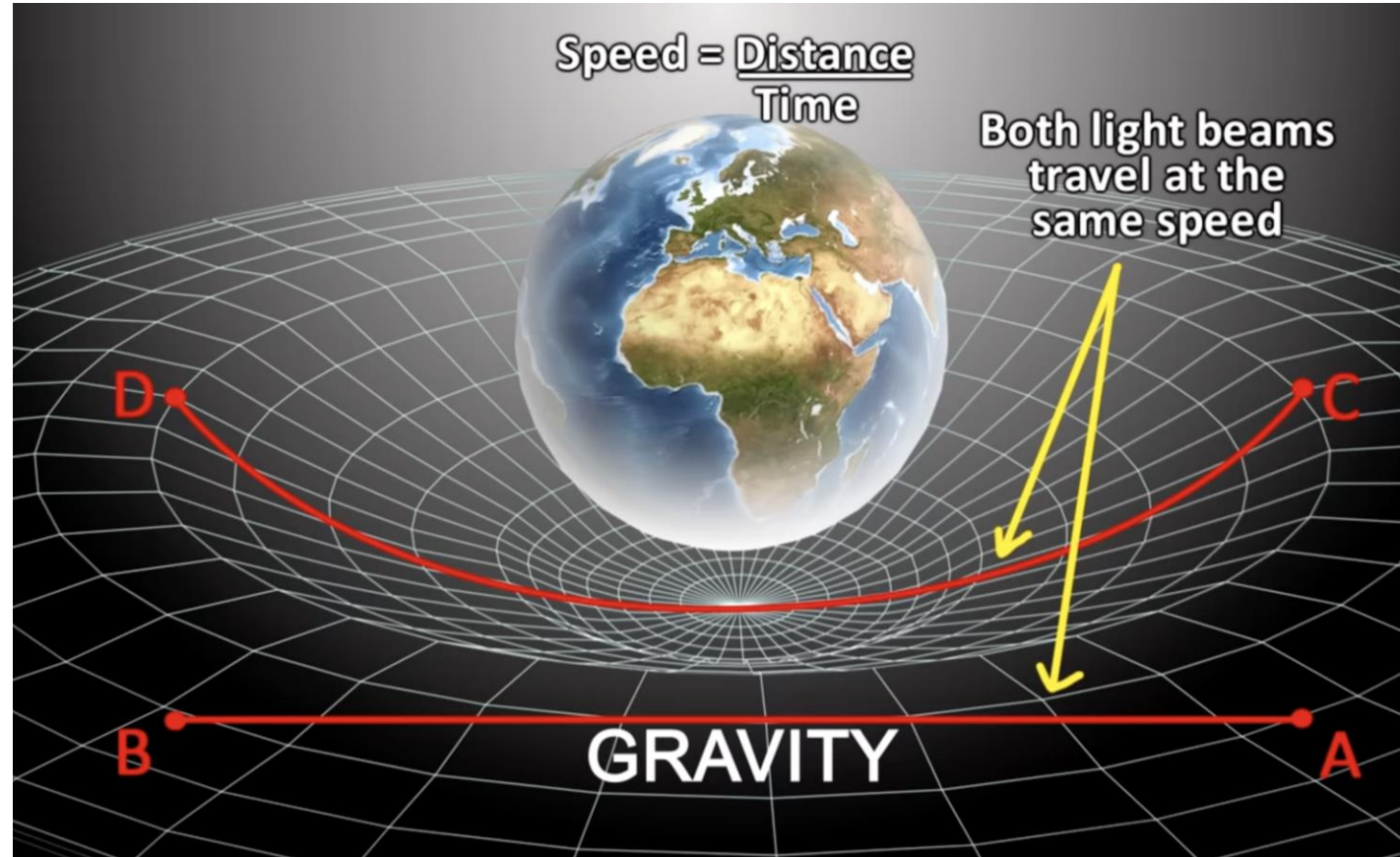




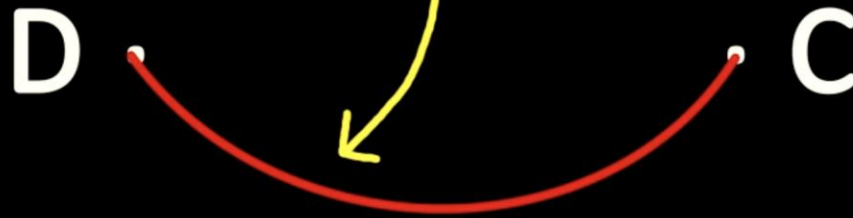
Einstein ring captured by Hubble telescope

Special Theory of Relativity

- As per Einstein : the speed of light is always the same between two points.
- But if one place gravity curves / distorts objects and at the other place it doesn't, How will the speed remain same?
- This is because gravity can change the time also.



Since distance is longer here, time must also be longer to keep a constant speed



constant
Speed = $\frac{\text{Distance}}{\text{Time}}$ Goes Up



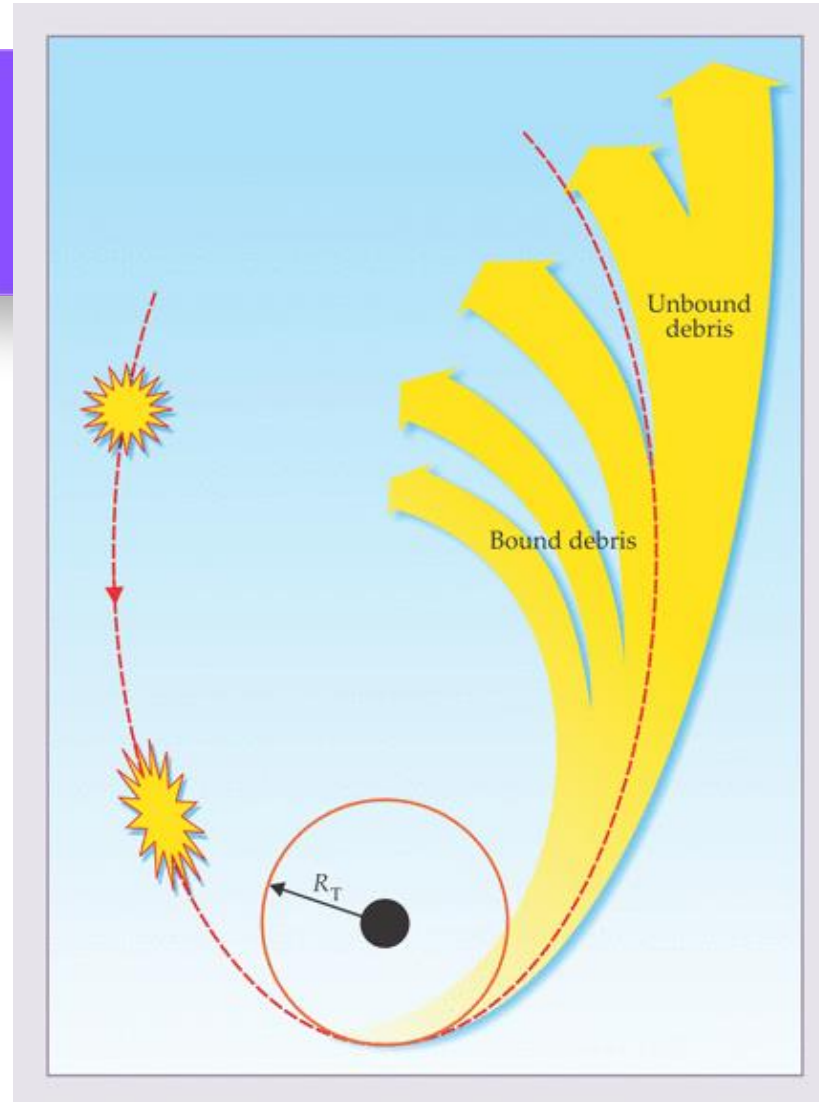
Conclusions

- Gravity is bending of space time : general theory of relativity
- Understanding black holes and how to photograph them
- Special theory of relativity : Space – time concept
- Gravitational Waves : in case of cataclysmic events in space

Tidal Disruption Events

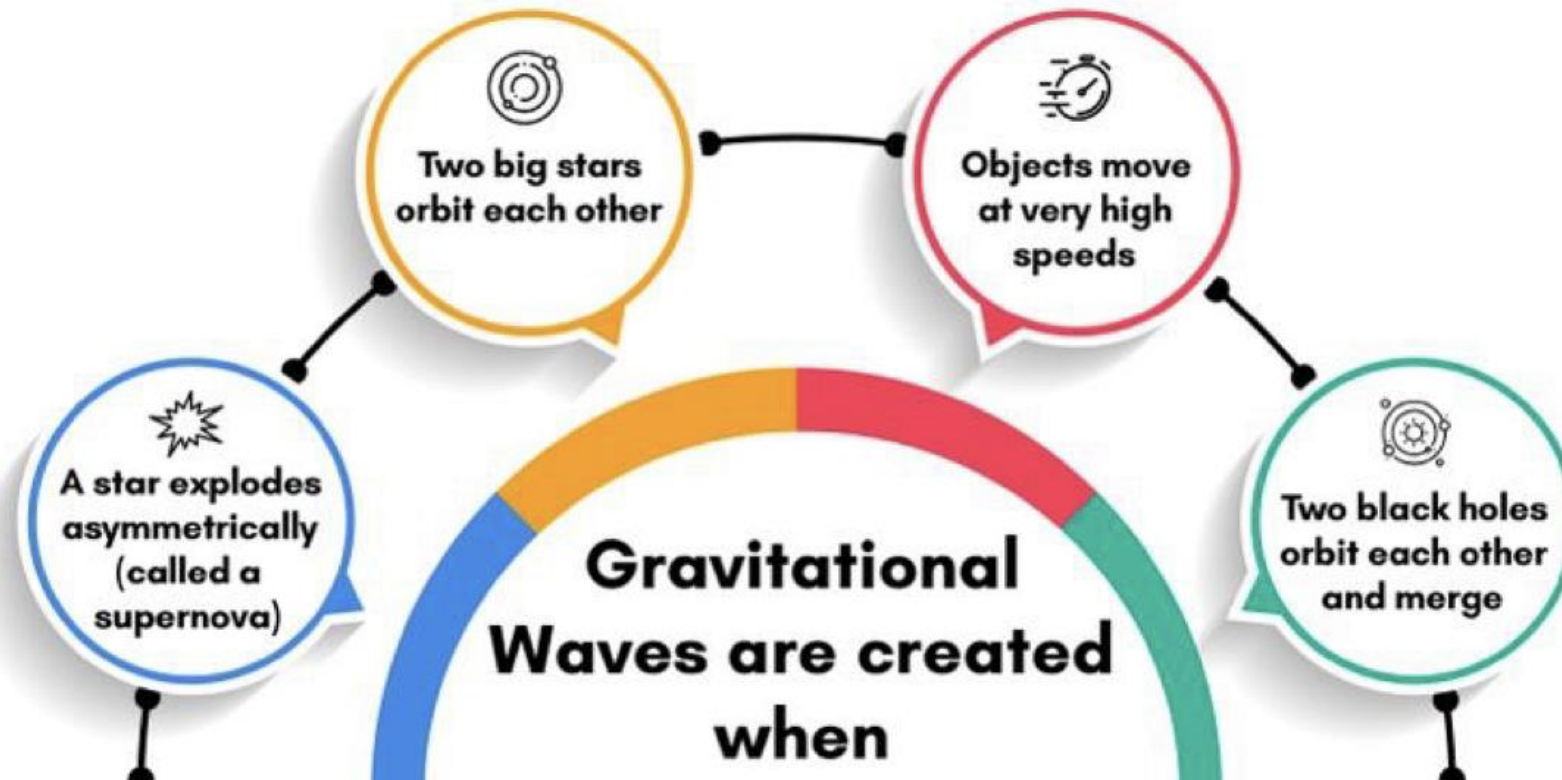
- 2018 : Scientists noticed that a black hole violently ripped apart a star when it got too close.
- Gravitational Forces stretch a star and then the star will be ripped apart.

Tidal Disruption Events



Binary Supermassive Black Hole

- Binary supermassive black hole system was discovered through gravitational lensing of blazars.
- Blazars : among the most luminous and energetic objects in the universe.
- Significance : will help in future detection of Graviational Waves



James Webb Space Telescope

- The James Webb Space Telescope (JWST) is NASA's infrared flagship observatory.
- It is an international collaboration between NASA, the European Space Agency (ESA), and the Canadian Space Agency (CSA).
- It was formerly called the Next Generation Space Telescope

James Webb Space Telescope

- Mission Goals are :
- Search for first galaxies or luminous objects formed after Big Bang.
- Big Bang is also called expanding universe hypothesis. It states that all of the current and past matter in the Universe came into existence at the same time, roughly 13.8 billion years ago.
- Determine how galaxies evolved.

James Webb Space Telescope

- Observe stars formation from the first stages to formation of planetary systems
- Measure physical and chemical properties of planetary systems, including our own Solar System, and investigate potential for life in those systems.

JWST continued

- JWST is headed for Sun-Earth Lagrange Point 2 (L2).
- The JWST is being considered as the successor of the Hubble Space Telescope (HST).

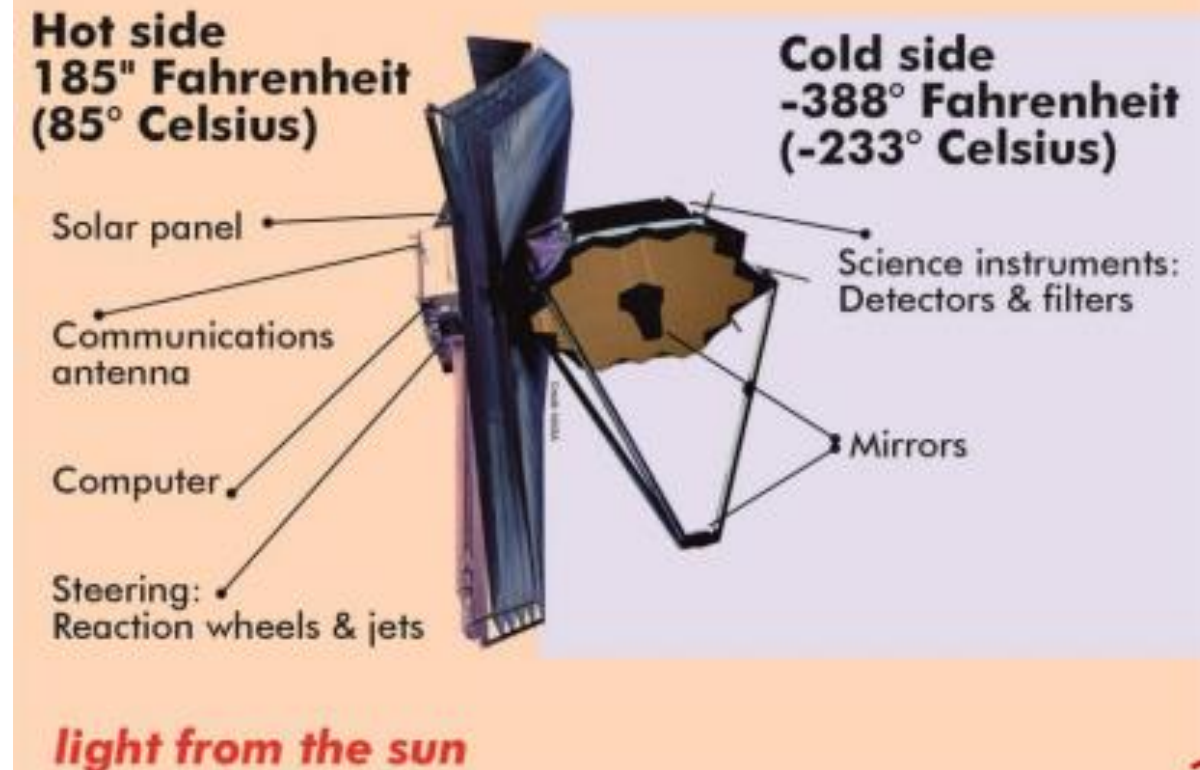
JSWT continued

- Advantages of point L2 :
- Spacecraft can communicate with earth
- Can observe the universe without chances of eclipse : clear view of deep space
- Solar power from the sun behind it

JAMES WEBB SPACE TELESCOPE

- James Webb Space Telescope has provided astronomers with a glimpse of the early universe in a new image
- Telescope captured an image of a galaxy cluster called MACS0647, as well as distant galaxy MACS0647-JD.
- Scientists believe that this might be an event of a galaxy merger in the early universe.
- The distant galaxy is visible because of gravitational lensing.

The Two Sides of the Webb Telescope



Key features of Webb

- Visibility Spectrum: Webb views the universe in infrared : **ONLY** specialized infrared telescope in space
- Mirrors: Its primary mirror is 6.5 meters in diameter
- A secondary mirror was also deployed
- L2 lets telescope stay in line with Earth as it moves around Sun.
- Allows satellite's large sunshield to protect telescope from light and heat of Sun and Earth

Major instruments:

- Contained within the Integrated Science Instrument Module (ISIM), which is one of the 3 major payloads.
- Optical Telescope Element (OTE) and Spacecraft Element are the other 2.
- ISIM is main payload. It houses four main instruments:
- Near-Infrared Camera
- Near-Infrared Spectrograph (NIRSpec),
- Mid-Infrared Instrument (MIRI) and
- Fine Guidance Sensor/ Near InfraRed Imager and Slit less Spectrograph (FGS/NIRISS)

How JWST will help in advancing understanding of Universe

Early Universe



JWST will allow us to see galaxies that formed just 100 million years after **Big Bang**

Exoplanets



JWST will scan atmospheres of **earth-like exoplanets**, searching for gases associated with life

Outer Planets



Will monitor Jupiter, Saturn, Uranus, and Neptune to see **how their atmospheres change** over time.

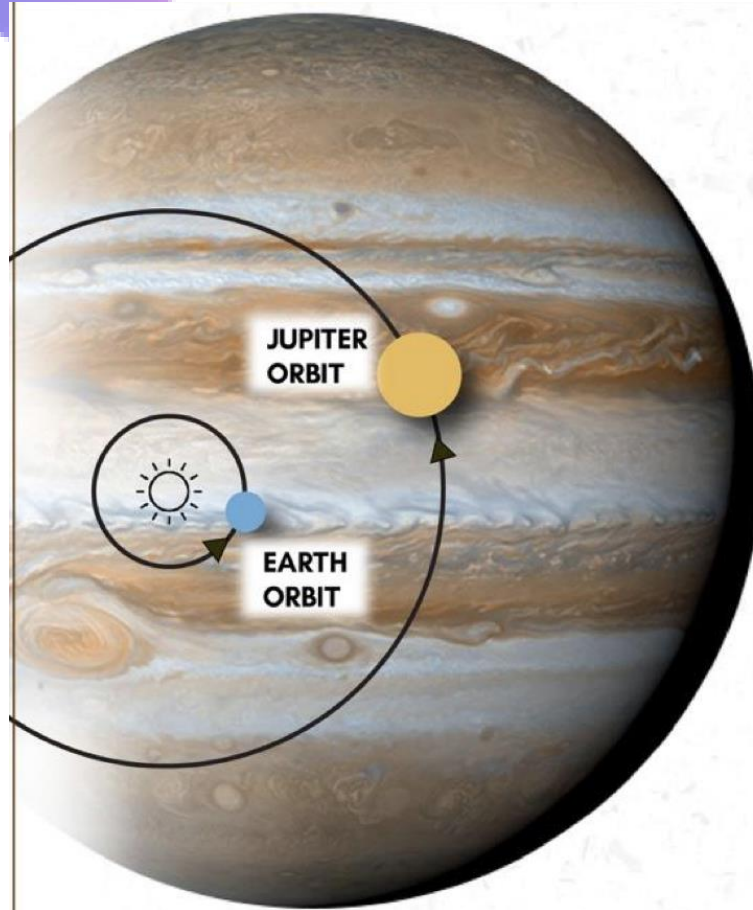
Small Worlds



From the **asteroid belt to Kuiper belt**, JWST will determine what makes up some of our solar system's smallest objects ★

OBSERVATIONS MADE BY JWST

- New images of Jupiter
- New images of Jupiter presented its massive storms, colourful auroras, faint rings and two small moons — Amalthea andAdrastea (Nearer to Jupiter).



JUPITER

588 TO 928
MILLION KILOMETRES

Distance from Jupiter to Earth depending on their orbits.

1. Rings

Unlike Saturn, Jupiter's rings are very faint and made of dust, not ice.

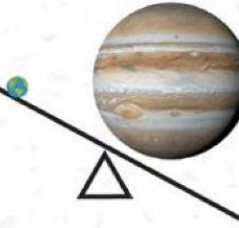


3. Great Red Spot

It has big storms like Great Red Spot, which has been going for hundreds of years. Recently, three smaller ovals merged to form Little Red Spot, about half the size of great red spot.



JUPITER
11 times
THE SIZE OF
EARTH



JUPITER
318 times
THE MASS OF
EARTH

2. Moons

Jupiter has maximum number of moons and its four largest moons - Io, Europa, Ganymede, and Callisto.



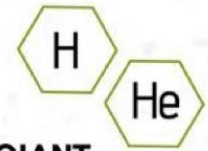
4. Auroras

Jupiter's magnetic field also causes aurorae at the planet's poles.



5. GAS GIANT

Made up of about 86% hydrogen and 14% helium, Jupiter is the largest planet orbiting our Sun, its volume is about 1.5 times greater than that of all the other planets in the solar system combined.



Pillars of Creation (PoC) – Formation of new stars

- (JWST) has captured the iconic PoC where new stars are forming within dense clouds of gas and dust.
- The three-dimensional pillars are made up of cool interstellar gas and dust that appear – at times – semi-transparent in near-infrared light
- Located 6,500 light years from Earth, in the Eagle Nebula of our Milky Way galaxy
- It will help researchers revamp their models of star formation



Exo-Moons

- Scientists at Indian Institute of Astrophysics have developed a model to trace habitable exomoons with the help of the James Webb Space Telescope (JWST)
- They are natural satellites that revolve around exoplanets (planets orbiting stars other than the sun)
- So far, 5000 exoplanets have been discovered by using several telescopes
- However, the natural satellites or exo-moons around any of these planets still remain untraced.

Other observations made from images revealed by Webb

- SMACS 0723 (Called Webb's First Deep Field): It is a cluster teeming with thousands of galaxies
- WASP-96b ; planet outside our solar system reveals clear signature of water,
- Southern Ring Nebula: This planetary nebula is approximately 2,000 light-years away.

Other observations made from images revealed by Webb

- Stephan's Quintet: galaxies, located in constellation Pegasus.
- Carina Nebula : “Cosmic Cliffs” in the Carina Nebula unveils the earliest, rapid phases of star formation
- Einstein Ring: n light from a star or a galaxy passes another galaxy or a massive object on its way towards Earth.

Term

- Kulper Belt : A donut shaped region of icy bodies beyond orbit of Neptune.
- Asteroid belt : A region of space between the orbits of mars and jupiter.
- Exoplanet : Any planet beyond our solar system
- Nebula : It is a giant cloud of dust and gas in space. Some nebulae come from gas and dust thrown out by explosion of a dying star, such as a supernova. Other nebulae are regions where new start are beginning to form

	Hubble	Webb	Herschel Space Observatory
Distance Earth	507 Km	1.5 Million Km at Second Sun-Earth Lagrange point	Second Sun-Earth Lagrange Point
Primary Mirror Diameter	2.4 Meter	5.5 Meter	3.5 Meter
Searching for	Young Galaxies (12.5)Billion year ago)	New Born Galaxies (13.5 Billion years ago)	Most actively star -forming galaxies
Serviceable	Yes	No	No
Wavelengths	Visible, UV, Part of near-infrared	Near and mid infrared	Far-infrared and submillimeter

Parker Space Probe

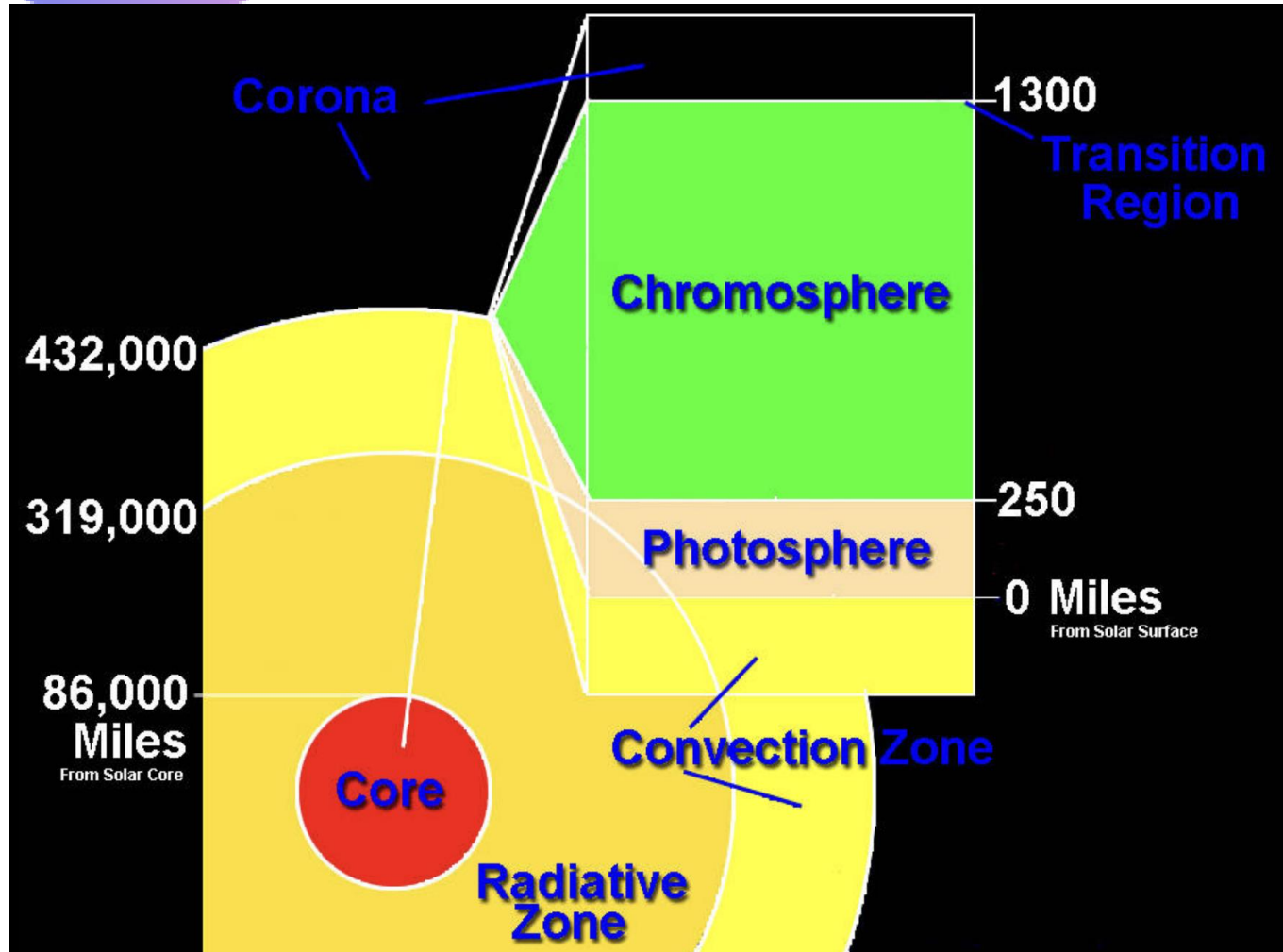
- NASA's Parker Solar Probe (PSP) has now flown through the Sun's upper atmosphere – the corona – and sampled particles and magnetic fields there.
- Launched in 2018, the PSP is part of NASA's Living with a Star program to explore aspects of the Sun-Earth system that directly affect life and society.

Parker Space Probe

- It is named after Eugene Newman Parker, a physicist who proposed concepts about how stars give off energy.
- It uses a carbon-composite shield, to withstand temperatures nearly 2,500 F (1,377 C), to protect itself from Sun's heat.

Parker Space Probe

- It became the first spacecraft in history to touch the Sun.
- It uses a series of gravitational encounters with the planet **Venus** to gradually lower the orbit of the spacecraft until it enters the outer atmosphere, or corona, of the Sun.



Mnemonic to remember layers of sun :

CoRaCoPh : Chrome and Corona

Relevance of PSP

- Deeper insights into sun's evolution.
- Predicting extreme weather events by helping in forecasting solar storms(or Geomagnetic storm).
- Solar storms are the biggest outbursts from the Sun can rattle Earth's magnetic field. In the process:
 - Communications may be disrupted,
 - Satellites can be knocked offline
 - Power grids will be vulnerable to electrical surges.

Solar Flares

- Solar flares are a sudden explosion of energy caused by reorganizing of magnetic field lines near sunspots.
- Solar magnetic cycle that works in deep interior of Sun creates regions that rise to surface and appear like dark spots. These are sunspots

Solar Flares

They appear dark because they are cooler than other parts of Sun's surface.

- In a solar flare, energy stored in sun's magnetic structures is converted into light and heat energy.
- This causes emission of high energy x-ray radiation and highly accelerated charged particles to leave the sun's surface.
- Sometimes solar flares also cause hot plasma to be ejected from Sun, causing a solar storm, and this is called Coronal Mass Ejection (CME).

Solar Flares

- Energy, radiation and high energy particles emitted by solar flares can trigger intense lights in the sky, called Auroras
- Auroras are caused by the interaction of energetic particles (electrons and protons) of the solar wind with atoms of the upper atmosphere occurring primarily in high latitudes of both hemispheres.
- Auroras in the Northern Hemisphere are called aurora borealis, aurora polaris, or northern lights, and in the Southern Hemisphere aurora australis, or southern lights.

Other Effects



Solar flare classification	Possible effects on Earth
B	none
C	Possible effects on space missions.
M	Blackout in radio transmissions and possible damages in astronauts outside spacecraft.
X	Damage to satellites, communication systems, power distribution stations and electronic equipment

Increasing Intensity



Solar Cycle

- Every 11 years, sun's magnetic field completely flips.
- This can be tracked using sunspots.
- Beginning of new solar cycle : Solar minimum.

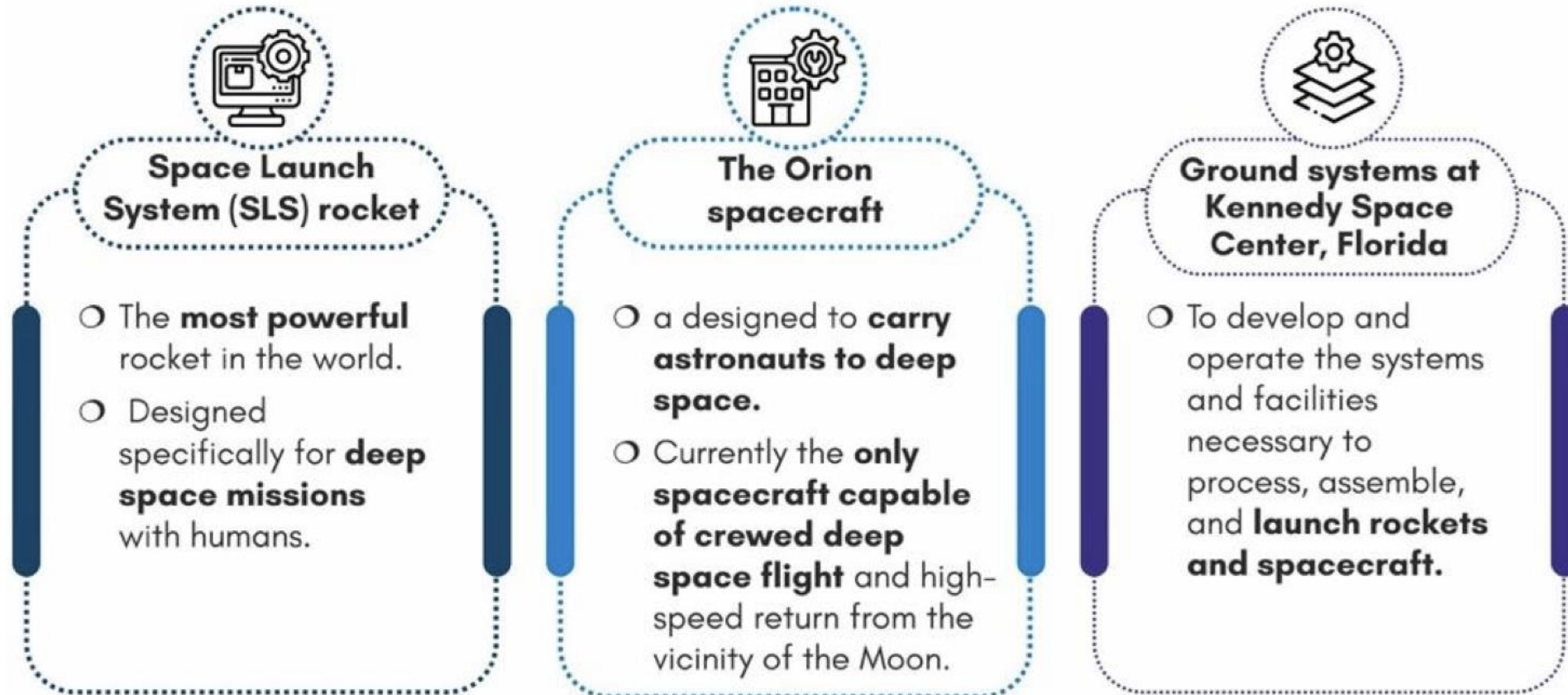
ARTEMIS I

- Recently, NASA's Artemis 1 mission successfully lifted off from the Kennedy Space Centre.
- Aim: To build a long term human presence at the Moon.

Mission specifications: An uncrewed flight test that will provide a foundation for human deep space exploration.

- With Artemis, NASA is planning to land the first woman on the Moon.
- First integrated test of NASA's deep space exploration systems,

Artemis 1



Artemis I Mission objectives

- Demonstrate Orion's heat shield can withstand the high speed only and high heat conditions when returning through Earth's atmosphere.
- Study the radiation environment of deep space
- Retrieving orion after splashdown
- Accomplish flight test objectives like certifying Orion's optical navigation system, deploying cube sats.
- Study the Moon to learn more about the origin and history Earth, the Moon, and our solar system.

USSR	• Luna 1, Luna 2, Luna 3. • Luna 2 was the first spacecraft to impact the Moon's surface.
USA	• Lunar Orbiter 1 • Apollo II: First humans to land on the moon. • Lunar Reconnaissance Orbiter (LRO)
Japan	• Hiten
China	• Change's 1
India	• Chandrayan - 1 • Chandrayaan-2

Retrograde orbit

- After getting closer to the moon, the Orion spacecraft used the gravitational kick it receives to enter a so-called "distant retrograde orbit."
- Retrograde means that it will orbit the moon in the opposite direction to the one in which the moon spins.
- Orion will stay in that orbit for some days. Then it will swing back down toward the moon for another kick to help power its journey back to Earth

WORD OF THE WEEK | PROGRADE vs RETROGRADE



The direction an object spins in relation to its orbit around the Sun. Prograde refers to an object that spins in the same direction as its orbit. Retrograde refers to an object that spins in the opposite direction of its orbit.

Earth has a prograde rotation and asteroid Bennu has a retrograde rotation, which means the two bodies rotate in opposite directions.



SPACE SUSTAINABILITY

- United Nations Office of Outer SPACE Affairs and government of AUSTRIA organized World Space Forum (WSF) 2022 under theme “Sustainability in Space for sustainability on Earth”.
- World Space Forum is a platform, to promote discussions
- Space sustainability
 - Ensuring that all humanity can continue to use outer space for peaceful purposes

Threats to space sustainability

- Orbital crowding and Space Debris:
- Militarisation and weaponisation of the space: Antisatellite (ASAT) weapons by countries like US, Russia, China, and India.
- Rendezvous and proximity operations (RPO): It involves altering the trajectory of one or more space objects to bring them into proximity to each other.
- Space weather can impact satellites

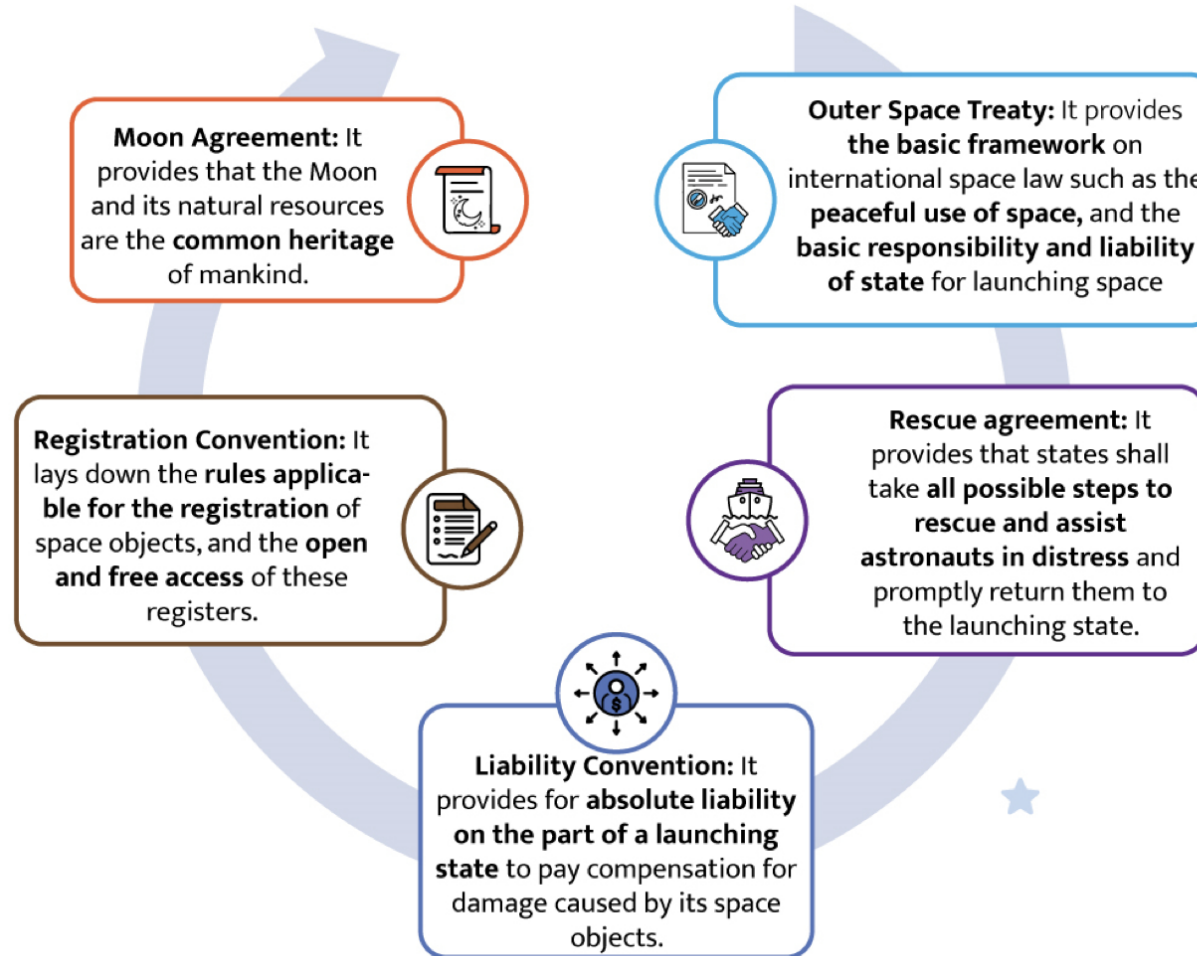
Initiatives taken towards space sustainability : Global

- U.N. Office for Outer Space Affairs (UNOOSA):
- Conference on Disarmament (CD): international forum with focus on prevention of an arms race in outer space (PAROS).
- Guidelines on space sustainability : by UNCOPOUS
- Space Sustainability Rating (SSR)
- ASAT test-ban resolution: A non-binding resolution (India abstained) passed by United Nations General Assembly (UNGA) calling for a ban on kinetic ASAT tests.

Indian Initiatives

- Space Situational Awareness : necessary mitigation measures to safeguard assets.
- It involves : SST : space surveillance and tracking
- Space Weather monitoring
- Near earth objects monitoring
- Project NETRA
- Digantra : space sector startup is going to set up India's first commercial SSA observatory.
- India USA bilateral SSA agreement.
- SPADEX : in orbit servicing : platform by ISRO

COPOUS oversees **five international treaties** underpinning space law:



Test of 13 Questions!

Cut off : 10/13



Q.1) PYQ (2013)

Q. The known forces of nature can be divided into four classes, viz, gravity, electromagnetism, weak nuclear force and strong nuclear force. With reference to them, which one of the following statements is not correct?

- a) Gravity is the strongest of the four
- b) Electromagnetism act only on particles with an electric charge
- c) Weak nuclear force causes radioactivity
- d) Strong nuclear force holds protons and neutrons inside the nuclear of an atom.

Q.2) PYQ (2016)

Q. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct?

1. Other than the USA and Russia, India is the only country to have launched a similar observatory into space.
2. Astrosat is a 2000 kg satellite placed in an orbit at 1650 km above the surface of the Earth.

Select the correct answer using the code given below

- A. 1 only

Q.3) PYQ (2016)

Q. Consider the following statements:

The Mangalyaan launched by ISRO

1. is also called the Mars Orbiter Mission
2. Made India the second country to have a spacecraft orbit Mars after the USA
3. Made India the only country to be successful in making its spacecraft orbits Mars in its very first attempt.

Which of the statements given above is/are correct?

Q.4) PYQ (2017)

Q. What is the purpose of 'evolved Laser Interferometer Space Antenna (eLISA)' project?

- a) To detect neutrinos
- b) To detect gravitational waves
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Q.5) PYQ (2017)

Q. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of

- A. Observation and understanding of the Universe
- B. Study of the solar and the lunar eclipses
- c. Placing satellites in the orbit of the Earth
- D. Origin and evolution of living organisms on the Earth

Q.6) PYQ (2018)

Q. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements :

1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

A. 1 only

Q.7) PYQ (2018)

Q. Consider the following phenomena :

1. Light is affected by gravity.
2. The Universe is constantly expanding.
3. Matter warps its surrounding space-time.

Which of the above is/are the prediction/predictions of Albert Einstein's General Theory of Relativity, often discussed in media?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 2
- d) 1, 2 and 3

Q.8) PYQ (2018)

Q. In which of the following areas can GPS technology be used?

1. Mobile phone operations
2. Banking operations
3. Controlling the power grids

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only

Q.9) PYQ (2018)

Q. With reference to India's satellite launch vehicles, consider the following statements:

1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-stage launch vehicle with the first and third stages using solid rocket motors, and the second and fourth stages using liquid rocket engines.

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 and 3
- C. 1 and 2
- D. 3 only

Q.10) PYQ (2019)

Q. Recently, scientists observed the merger of giant ‘blackholes’ billions of light-years away from the Earth. What is the significance of this observation?

- a) ‘Higgs boson particles’ were detected.
- b) ‘Gravitational waves’ were detected.
- c) Possibility of intergalactic space travel through ‘wormhole’ was confirmed.
- d) It enabled the scientists to understand ‘singularity’.

Q.11) PYQ (2020)

Q. “The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft”. The experiment in question refers to

- a) Voyager
- b) New Horizons
- c) LISA Pathfinder
- d) Evolved LISA

Q.12) PYQ (2021)

Q. Which one of the following is a reason why astronomical distances are measured in light-years?

- a) Distances among stellar bodies do not change.
- b) Gravity of stellar bodies does not change.
- c) Light always travels in a straight line.
- d) Speed of light is always the same.

Q.13) PYQ (2022)

Q. If a major solar storm (solar flare) reaches the Earth, which of the following are the possible effects on the Earth?

1. GPS and navigation systems could fail.
2. Tsunamis could occur at equatorial regions.
3. Power grids could be damaged.
4. Intense auroras could occur over much of the Earth.
5. Forest fires could take place over much of the planet.
6. Orbits of the satellites could be disturbed.
7. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

Select the correct answer using the code given below:

- | | |
|---------------------------|----------------------------|
| (a) 1, 2, 4 and 5 only | (b) 2, 3, 5, 6 and 7 only |
| (c) 1, 3, 4, 6 and 7 only | (d) 1, 2, 3, 4, 5, 6 and 7 |

Congratulations on giving the test!

Let's Discuss The Answers.



Q.1) PYQ (2013)

Q. The known forces of nature can be divided into four classes, viz, gravity, electromagnetism, weak nuclear force and strong nuclear force. With reference to them, which one of the following statements is not correct?

- a) Gravity is the strongest of the four**
- b) Electromagnetism act only on particles with an electric charge
- c) Weak nuclear force causes radioactivity
- d) Strong nuclear force holds protons and neutrons inside the nuclear of an atom.

Answer: A

- Statement 1 is incorrect since gravity is actually the weakest of the four fundamental forces for a given distance
- Electromagnetism acts only on charged particles, where the charge can be negative or positive
- The weak force is responsible for the existence and structure of atomic nuclei including radioactivity and nuclear fusion
- Strong nuclear force is responsible for holding protons and neutrons together inside the atomic nucleus and is 100 times stronger than electromagnetism

Q.2) PYQ (2016)

Q. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct?

1. Other than the USA and Russia, India is the only country to have launched a similar observatory into space.
2. Astrosat is a 2000 kg satellite placed in an orbit at 1650 km above the surface of the Earth.

Select the correct answer using the code given below

- A. 1 only

Answer: D

- ASTROSAT is **India's first dedicated multi-wavelength space observatory.**
- This scientific satellite mission endeavours for a more **detailed understanding of our universe.**
- **Before India, NASA, European Space Agency, JAXA, Roscosmos have launched similar observatory into the space.**
- **ASTROSAT with a lift-off mass of about 1513 kg was launched into a 650 km orbit inclined at an angle of 6 deg to the equator by PSLV-C30.**

Q.3) PYQ (2016)

Q. Consider the following statements:

The Mangalyaan launched by ISRO

1. is also called the Mars Orbiter Mission
2. Made India the second country to have a spacecraft orbit Mars after the USA
3. Made India the only country to be successful in making its spacecraft orbits Mars in its very first attempt.

Which of the statements given above is/are correct?

Answer: D

- The Mars Orbiter Mission (MOM), also called **Mangalyaan is a space probe orbiting Mars since 24 September 2014.**
- It was launched on 5 November 2013 **by the Indian Space Research Organisation (ISRO).**
- It is **India's first interplanetary mission** and it made it the fourth space agency to reach Mars, after Roscosmos, NASA, and the European Space Agency.
- It is the **first Asian nation to reach Mars orbit**, and the first nation in the world to do so in its first attempt

Q.4) PYQ (2017)

Q. What is the purpose of 'evolved Laser Interferometer Space Antenna (eLISA)' project?

- a) To detect neutrinos
- b) To detect gravitational waves**
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Answer: B

- The Laser Interferometer Space Antenna (LISA) is a European Space Agency mission designed **to detect and accurately measure gravitational waves—tiny ripples in the fabric of space-time**—from astronomical sources. LISA would be the first dedicated space-based gravitational wave detector. It aims to measure gravitational waves directly by using laser interferometry.

Q.5) PYQ (2017)

Q. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of

- A. Observation and understanding of the Universe**
- B. Study of the solar and the lunar eclipses
- c. Placing satellites in the orbit of the Earth
- D. Origin and evolution of living organisms on the Earth

Answer: A

- An **event horizon is most commonly associated with black holes.**
- Light emitted from inside the event horizon can never reach the outside observer.
- In physics, **string theory is a theoretical framework in which the point-like particles of particle physics** are replaced by one-dimensional objects called strings.
- It describes how these strings propagate through space and interact with each other

Q.6) PYQ (2018)

Q. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements :

1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

A. 1 only

Answer: A

- The IRNSS is being developed parallel to the GAGAN (GPS Aided GEO Augmented Satellite Navigation) program, the ISRO SBAS (Satellite Based Augmentation System) version of an overlay system for GNSS signal corrections.
- The IRNSS system consists of a constellation of seven satellites and a supporting ground segment.
- Three of the satellites in the constellation will be placed in a geostationary orbit and the remaining four in a geosynchronous

Q.7) PYQ (2018)

Q. Consider the following phenomena :

1. Light is affected by gravity.
2. The Universe is constantly expanding.
3. Matter warps its surrounding space-time.

Which of the above is/are the prediction/predictions of Albert Einstein's General Theory of Relativity, often discussed in media?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 2
- d) 1, 2 and 3**

Answer: D

- 2010: International Team of Astronomers confirmed Albert Einstein's theory of general relativity and that the expansion of the universe is accelerating after looking at data from the Hubble Space Telescope. There is an unknown source of energy in the universe which is causing the cosmic expansion to speed up.
- 2016: Japanese researchers' study finds that the expansion of the universe could be explained by a cosmological constant, as proposed by Einstein in his theory of general relativity.

Q.8) PYQ (2018)

Q. In which of the following areas can GPS technology be used?

1. Mobile phone operations
2. Banking operations
3. Controlling the power grids

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3**

Answer: D

- GPS is the Global Positioning System.
- It's a global navigation satellite system that uses at least 24 satellites, a receiver and algorithms to provide location, velocity and time synchronization for air, sea and land travel. GPS works at all times and in almost all weather conditions.
- Some examples of GPS applications include Agriculture, Astronomy, Automated vehicle, Cellular telephony, Clock synchronization, Disaster relief/emergency services, Radio occultation for weather and atmospheric science applications, Geofencing, GPS aircraft tracking, for mining, or for data mining, for navigation and surveying.

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1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-stage launch vehicle with the first and third

Answer: A

- Polar Satellite Launch Vehicle (PSLV) is the third generation launch vehicle of India. It is the first Indian launch vehicle to be equipped with liquid stages.
- GSLV-Mk III is a three-stage vehicle with two solid motor strap-ons (S200), a liquid propellant core stage (L110) and a cryogenic stage (C25).
- Geostationary satellites orbit around the earth in 24 hours and since the earth rotates with the same period, the satellite would appear fixed from any point on earth.
- GSLV-Mk III is capable of launching 4 ton class of satellites to Geosynchronous Transfer orbit (GTO).

Q.10) PYQ (2019)

Q. Recently, scientists observed the merger of giant ‘blackholes’ billions of light-years away from the Earth. What is the significance of this observation?

a) ‘Higgs boson particles’ were detected.

b) ‘Gravitational waves’ were detected.

c) Possibility of intergalactic space travel through ‘wormhole’ was confirmed.

d) It enabled the scientists to understand ‘singularity’.

Answer: B

- It is possible for two black holes to collide.
- Once they come so close that they cannot escape each other's gravity, they will merge to become one bigger black hole.
- Such an event would be extremely violent.
- Even when simulating this event on powerful computers, we cannot fully understand it. However, we do know that a black hole merger would produce tremendous energy and send massive ripples through the space-time fabric of the Universe.
- These ripples are called gravitational waves.

Q.11) PYQ (2020)

Q. “The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft”. The experiment in question refers to

- a) Voyager
- b) New Horizons
- c) LISA Pathfinder
- d) Evolved LISA**

Answer: D

- Evolved Laser Interferometer Space Antenna (eLISA) is a spectacular plan of setting into space three spacecrafts, a mother and two daughter spacecrafts, which will fly in a triangular formation, trailing the earth in its orbit around the sun at a distance of over.
- 50 million km. Each arm of the imaginary triangle, from the mother to each daughter spacecraft, will measure about a million km.
- ELISA seeks to measure gravitational waves in the frequency range from 0.1 mHz to about 100 mHz. To achieve this, it is necessary for the interferometers to have an arm length of a million kilometres and that is impossible to achieve with an earth based setup.
- Therefore, option (d) is the correct answer.

Q.12) PYQ (2021)

Q. Which one of the following is a reason why astronomical distances are measured in light-years?

- a) Distances among stellar bodies do not change.
- b) Gravity of stellar bodies does not change.
- c) Light always travels in a straight line.
- d) **Speed of light is always the same.**

Answer: D

- A light-year is the distance a beam of light travels in a single Earth year, or 6 trillion miles (9.7 trillion kilometers).
- The reason for using light years to measure distance in space is because throughout the universe, light travels at exactly the same speed.
- Therefore, option (d) is the correct answer

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1. GPS and navigation systems could fail.
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7. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

Select the correct answer using the code given below:

- (a) 1, 2, 4 and 5 only (b) 2, 3, 5, 6 and 7 only
(c) 1, 3, 4, 6 and 7 only (d) 1, 2, 3, 4, 5, 6 and 7

Answer: C

- Solar flares are large explosions from the surface of the sun that emit intense bursts of electromagnetic radiation.
- Solar flares occur when magnetic energy builds up in the solar atmosphere and is released suddenly. These outbursts are intrinsically linked to the solar cycle — an approximately 11-year cycle of solar activity driven by the sun's magnetic field.
- They can lead to failing of GPS and navigation systems and could damage power grids on earth. They could also result in intense auroras over much of the Earth. High intensity solar flares also could disturb space activities.